

VPT-NSP²++: Importance-Aware Visual Prompt Tuning in Null Space for Continual Learning

Shizhou Zhang[✉], Yue Lu[✉], De Cheng[✉], Yinghui Xing[✉], *Member, IEEE*, Nannan Wang[✉], *Senior Member, IEEE*, Peng Wang[✉], *Member, IEEE*, and Yanning Zhang[✉], *Fellow, IEEE*

Abstract—Continual learning (CL) enables AI models to adapt to evolving environments while mitigating catastrophic forgetting, which is a critical capability for dynamic real-world applications. With the growing popularity of pre-trained Vision Transformer (ViT) models and visual prompt tuning (VPT) technique in CL, this work explores a CL method on top of the ViT-based foundation model, through VPT mechanism with theoretical guarantees. Inspired by the orthogonal projection method, we aim to leverage this approach for VPT to enhance CL performance, particularly in long-term scenarios. However, since the orthogonal projection is originally designed for linear operations in CNNs, applying it to ViTs poses challenges induced by the non-linear self-attention mechanism and the distribution drift within LayerNorm. To address these issues, we deduced two orthogonality conditions to achieve the prompt gradient orthogonal projection, which provide a theoretical guarantee of maintaining stability. Considering the strict orthogonal constraints can diminish model capacity and reduce plasticity, we further propose an importance-aware orthogonal regularization framework. By applying varying degrees of orthogonal constraints to different parameters based on their importance to old and new tasks, the framework adaptively enhances model capacity and thereby promotes long-sequence CL while improving the stability-plasticity trade-off. To implement the proposed approach, a null-space-based approximation solution is employed to efficiently achieve the prompt gradient orthogonal projection. Extensive experiments on various class-incremental learning benchmarks demonstrate that our method achieves state-of-the-art performance across diverse CL scenarios.

Index Terms—Continual learning, catastrophic forgetting, visual prompt tuning, null space, orthogonal projection.

I. INTRODUCTION

CONTINUAL Learning (CL) aims to enable AI models to continuously learn new tasks or data over time while

retaining knowledge from previously learned tasks. This capability is particularly valuable in dynamic environments such as robotics and autonomous driving, where AI systems must continuously adapt to new challenges and changing environments. In contrast to conventional AI paradigms which build static models tailored to fixed datasets, continual learning offers a dynamic framework that empowers models to evolve with ever-changing environments [1]. However, a key challenge in continual learning is *catastrophic forgetting*, where models tend to lose previously acquired knowledge when they learn new tasks.

To address catastrophic forgetting, existing continual learning methods with CNN-based architectures primarily rely on three techniques: rehearsal-based approaches [2], [3], [4], network expansion-based methods [5], [6], [7], and regularization-based strategies [8], [9], [10]. While these methods have made significant progress in CNNs, large-scale pre-trained Vision Transformers (ViTs) [11] have gained increasing attention due to their superior generalizability and performance across a wide range of vision tasks [11], [12], [13]. In ViT-based CL, parameter-efficient fine-tuning techniques [14], [15], [16] are commonly used to adapt pre-trained ViTs to new tasks. Among these, visual prompt tuning (VPT) [14] has emerged as a prominent approach [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], owing to its ability to introduce minimal task-specific parameters while preserving the representation capacity of the pre-trained model. Given its superiority, this work explores the possibility of building a CL method on top of the ViT-based foundation model, through VPT mechanism with theoretical guarantees.

In CNN-based CL, *orthogonal projection* approaches [10] offer a theoretical solution to address catastrophic forgetting through updating convolutional kernels in the direction orthogonal to the subspace of previous inputs. Inspired by the success of these approaches, we aim to leverage orthogonal projection for prompt tuning in ViT-based foundation models to enhance continual learning performance, especially in long-term CL scenarios. This approach maintains consistent representations on previous tasks even as the prompts are adapted to the new task, providing theoretical guarantees for updating prompts without forgetting. However, as the orthogonal projection in CNNs is designed for linear operations, directly applying it to the ViT-based foundation model through prompt tuning technique still presents several challenges: 1) The self-attention [30] operation in ViT module is high-order and non-linear. 2) The

Received 11 January 2025; revised 11 July 2025; accepted 6 December 2025. Date of publication 11 December 2025; date of current version 6 March 2026. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 62576282, Grant 62576262, Grant 62476223, and Grant 62176198, and in part by the Innovation Capability Support Program of Shaanxi under Program 2024ZC-KJXX-043. Recommended for acceptance by L. Jiang. (Shizhou Zhang and Yue Lu are co-first authors.) (Corresponding author: De Cheng.)

Shizhou Zhang, Yue Lu, Yinghui Xing, Peng Wang, and Yanning Zhang are with the School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: szzhang@nwpu.edu.cn; zgxd@mail.nwpu.edu.cn; xyh_7491@nwpu.edu.cn; peng.wang@nwpu.edu.cn; ynzhang@nwpu.edu.cn).

De Cheng and Nannan Wang are with the School of Telecommunications Engineering, Xidian University, Xi'an 710126, China (e-mail: dcheng@xidian.edu.cn; nnwang@xidian.edu.cn).

Digital Object Identifier 10.1109/TPAMI.2025.3642298

LayerNorm [31] preceding self-attention induces a drift in the prompt distribution. Specifically, the orthogonal projection for CNNs is drawn from linear operations, where the input feature matrix is directly multiplied by the weight matrix. In contrast, the self-attention mechanism in Transformers involves not only three linear transformations in parallel, but also higher-order interactions of tokens and non-linear activations on the attention map. These factors render the relationship between prompts and representations in ViTs much more complex, far surpassing the linearity in CNNs.

To address these challenges, we theoretically deduced two orthogonality conditions to achieve the prompt gradient orthogonal projection, which provide a theoretical guarantee of maintaining consistent representations via the self-attention mechanism in visual prompt tuning. To be specific, we firstly take the full self-attention and LayerNorm into consideration and derive a strict condition through a comprehensive analysis of the forward propagation of the ViT layer. Subsequently, we propose to reformulate the condition for self-attention into its two sufficient conditions, which enables us to address the challenge of high order and nonlinearity. Following, we propose a constraint of invariant prompt distribution that removes the obstacle to the final simplification of the conditions brought by the LayerNorm. Ultimately, we derive two sufficient orthogonality conditions, which indicate that if the prompt update lies in the subspaces orthogonal to: 1) the normalized old image tokens multiplied by the qkv-transformation matrices, and 2) the activated attention map computed by the old image queries and prompt keys, then the prompt updates can maintain consistent representations and thus guarantee stability theoretically. In reality, the proposed orthogonal subspace constraint can also be directly applied to the full fine-tuning setting for CL in the ViT-based foundation model.

Although the strict orthogonal constraints ensure stability, they diminish model capacity and consequently limit plasticity. Given that different parameters contribute unequally to learning old tasks and adapting to new tasks, we eventually propose an importance-aware orthogonal regularization framework to adaptively enhance model capacity, which is particularly beneficial for long-sequence CL. Specifically, based on the dimensional importance of prompts to both old and new tasks, we partition the prompt dimensions into four groups and apply different (more relaxed or stricter) degrees of orthogonal constraints. During training on the new task, a strict orthogonal constraint is performed on the parameters crucial for both old and new tasks. Parameters essential only for the new task are allowed to update without orthogonal constraints, while those solely critical to the old task keep frozen. For the remaining parameters with moderate importance to both old and new tasks, a relaxed orthogonal constraint is applied. By doing this, we can balance between stability and plasticity in a flexible and adaptive manner, hence improving overall performance.

In practice, we utilize a null-space-based approximation solution [32] to implement the prompt gradient orthogonal projection. We validate our proposed approach, VPT-NSP²⁺⁺, on various class-incremental learning benchmarks, demonstrating

its effectiveness and superiority in different continual learning settings.

We summarize our contributions as follows:

- We completely introduce the orthogonal projection into visual prompt tuning for ViT-based continual learning to address catastrophic forgetting theoretically. To this end, two sufficient consistency conditions for the self-attention and an invariant prompt distribution constraint for LayerNorm are derived, which ensures representation consistency during prompt updates and thereby provides theoretical guarantees on maintaining stability.
- As strict orthogonal constraints diminish model capacity and hinder plasticity, we further propose an importance-aware orthogonal regularization framework to address this limitation. It adaptively enhances model capacity by applying varying degrees of orthogonal constraints to different parameters, based on their respective importance to old and new tasks, hence promoting long-sequence CL and improving the stability-plasticity trade-off.
- Extensive experiments on various benchmarks demonstrate that our approach outperforms existing state-of-the-art methods by a large margin, particularly in challenging long-sequence CL scenarios.

This paper is an extension of our conference paper [33] published in NeurIPS 2024. The differences and improvements lie in the following aspects: (1) We propose an importance-aware orthogonal regularization framework that applies distinct orthogonal constraint strategies on different prompt dimensions, thereby enhancing the stability-plasticity balance. (2) A classifier alignment technique is seamlessly integrated into our framework, where it can maximize its effectiveness in mitigating inter-task confusion. (3) Additional experiments are conducted in long-sequence and out-of-distribution continual learning scenarios. Results demonstrate the substantial improvements and generalizability of our approach compared to our previous work [33] and existing state of the arts.

II. RELATED WORK

A. CNN-based Continual Learning

CNN-based CL methods are typically trained from scratch without relying on large-scale pre-training [34], [35], [36]. There are three primary categories of CNN-based CL methods: 1) The rehearsal-based methods [2], [3], [4] store a subset of training samples and use them in conjunction with new samples to train for subsequent tasks. The critical challenge in these methods lies in determining which samples to preserve to effectively retain prior knowledge. For instance, iCaRL [2] employs a herding strategy to ensure that the prototype (*i.e.*, average feature vector) of the stored samples in the feature space closely approximates the prototype of all training samples. 2) The network expansion-based methods [5], [6], [7] retain parameter branches from previous tasks and expand the network by adding a new branch for each new task. For example, DER [5] freezes the previously learned feature extractor upon encountering a new task and extends the backbone with a new feature extractor. The

features from all extractors are then concatenated for classification. However, the expansion of branches leads to increased inference overhead. 3) The regularization-based methods [8], [9], [10] constrain parameter updates to prevent significant changes during incremental learning. As an example, EWC [8] associates the constraint of parameters with the loss function, enabling it to identify balancing parameters that can solve new tasks without causing significant performance degradation on old tasks. Another subcategory of regularization-based methods is orthogonal projection, which will be discussed later.

B. Pre-trained Transformer-based Continual Learning

Benefited from the development of parameter-efficient tuning techniques [14], [15], [16], [37], [38], [39], [40], continual learning based on pre-trained large-scale Transformers has gained widely attention in recent years. Existing works can be broadly categorized into the following four types. 1) *Prompt pool-based methods*: The mainstream works firstly query instance-relevant prompt from a prompt pool and then feed the retrieved prompt into the pre-trained ViT [11] for a further classification [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]. For instance, L2P [17] proposes a shared prompt pool to store the key-prompt pairs across tasks. On top of L2P, DualPrompt [18] adds expert prompts for each task. CODA-Prompt [20] proposes weighted prompt selection and fusion, enabling the framework trainable in an end-to-end manner. ConvPrompt [29] dynamically generates task-specific prompts by convolving the task-shared parameters. 2) *Slow updating-based methods*: An intuitive strategy to alleviate forgetting is slowly updating parameters [41], [42]. For instance, LAE [41] trains an online fine-tuning module on new tasks, and gradually accumulates the learned knowledge into an offline fine-tuning module through momentum updates. 3) *Branch expansion-based methods*: Inspired by continual learning in CNNs, some methods introduce expandable branches into pre-trained ViTs [43], [44], [45], [46]. For example, based on a structured attention [47] that decouples the interaction of prompts, APT [43] learns task-specific prompts independently for each task, and combines all prompts for inference. 4) *Classifier enhancement-based methods*: A line of works focusing on learning discriminative classifiers across tasks [48], [49], [50]. As an example, ESN [48] constrains the energy value of each task-specific classifier around a fixed energy anchor to ensure the alignment among multiple task classifiers.

Although existing methods attempt to reduce forgetting via various techniques, the change in representations learned from previous instances remains inevitable, gradually resulting in catastrophic forgetting.

C. Orthogonal Projection in Continual Learning

A line of CNN-based continual learning methods [10], [32], [51], [52], [53], [54] have revealed that during the learning of a new task, if the weights of convolutional or fully-connected layers are updated in the direction orthogonal to the input features of previously learned instances, the output representations of these instances remain unaffected by the weight updates. Therefore, forgetting can be prevented in theory. To achieve orthogonality

between weight updates and the features of old inputs, three main techniques are employed: 1) *Correlation-inverse matrix*: For example, OWM [10] constructs a projector for gradients by calculating the correlation-inverse matrix [55]. 2) *Subtraction of feature space* [52], [56], [57]: To avoid matrix inversion, GPM [52] firstly projects the gradients into the subspace spanned by input features, and then subtracts the projected components to achieve orthogonality. 3) *Null space* [32], [53], [58]: Several studies leverage the orthogonal relationship between the feature space and its null space [59]. For instance, Adam-NSCL [32] projects weight updates onto the approximate null space of previous input features, offering a more straightforward approach. The primary difference among the aforementioned methods lies in their implementation of orthogonal projection. They share a common orthogonality condition, *i.e.*, the weight updates should be orthogonal to the subspace spanned by previous input features.

Nonetheless, the orthogonality condition formulated for CNNs is not directly applicable to prompt tuning within ViTs. In this study, we address this issue by formulating orthogonality conditions specific to prompt tuning.

III. PRELIMINARIES

A. Continual Learning

Under the protocol of continual learning, a model is trained over a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T$. In the t -th task $\mathcal{T}_t$, the training set is denoted as $\mathcal{D}_t = \{x_t^{(n)}, y_t^{(n)}\}_{n=1}^{|\mathcal{T}_t|}$, where $x_t^{(n)}$ is the n -th image with label $y_t^{(n)}$, and $|\mathcal{T}_t|$ denotes the number of instances in this task. We focus on the class-incremental learning scenario [60], [61], [62]: the label space $\mathcal{Y}_t$ of task $\mathcal{T}_t$ is disjoint with other tasks, *i.e.*, $\mathcal{Y}_t \cap \mathcal{Y}_{t'} = \emptyset$ for $t \neq t'$. Once the model completes learning on task $\mathcal{T}_t$, the corresponding training set $\mathcal{D}_t$ will become unavailable when learning from $\mathcal{T}_{t+1}$. After sequential training, the model is expected to accurately classify test samples from all previously learned classes without requiring task identification.

B. Visual Prompt Tuning

The forward propagation of visual prompt tuning is illustrated in Fig. 1. For an instance x , whose task identity and sample index are omitted, its N image tokens (including the class token) to be fed into a ViT layer $f_{\text{vit}}(\cdot)$ (with the layer index neglected) are represented as $\mathbf{X} \in \mathbb{R}^{N \times D}$, where D denotes the dimension of each token. In prompt tuning, the prompts $\mathbf{P} \in \mathbb{R}^{M \times D}$ of length M are fed into the ViT layer along with the image tokens $\mathbf{X}$, and their concatenation is denoted as $\mathbf{Z} = [\mathbf{X}; \mathbf{P}] \in \mathbb{R}^{(N+M) \times D}$ for simplicity. All the tokens firstly undergo the LayerNorm [31] (denoted as $\text{LN}(\cdot)$) to calibrate the distributions:

$$\text{LN}(\mathbf{Z}) = \frac{\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}}}{\boldsymbol{\sigma}_{\mathbf{Z}}} \odot \boldsymbol{\alpha} + \boldsymbol{\beta}, \quad (1)$$

where $\boldsymbol{\mu}_{\mathbf{Z}}, \boldsymbol{\sigma}_{\mathbf{Z}} \in \mathbb{R}^{N+M}$, $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}^D$ are vectors and can be broadcast to matrices of dimensions $(N+M) \times D$. The symbol $\odot$ and division represent the element-wise product and division, respectively.

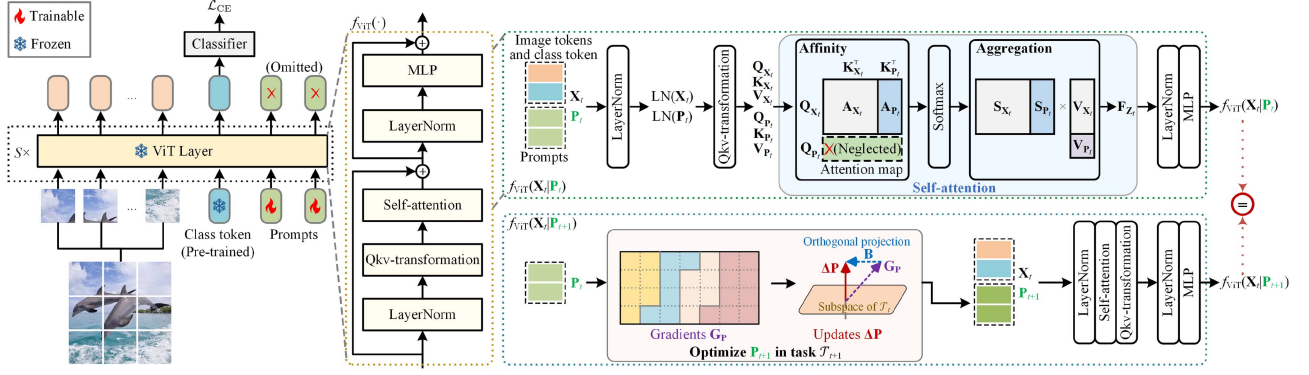


Fig. 1. Framework of our method. Trainable prompts $\mathbf{P}$ are incorporated into ViT layers, while other parameters are pre-trained and frozen. During training in the new task $\mathcal{T}_{t+1}$, the gradients of the prompts, $\mathbf{G}_\mathbf{P}$, are projected onto the orthogonal subspace of the previous task $\mathcal{T}_t$, with different orthogonal constraints applied based on their importance. Consequently, the updated prompts $\mathbf{P}_{t+1}$ (adjusted by $\Delta\mathbf{P}$) can keep the output of the previous instance ($\mathbf{X}_t$) unchanged to effectively mitigate forgetting.

Each normalized token is then processed through the qkv-transformations, which consist of three parallel linear layers:

$$\mathbf{Q}_\mathbf{Z} = \text{LN}(\mathbf{Z})\mathbf{W}_q + \mathbf{b}_q, \quad (2)$$

$$\mathbf{K}_\mathbf{Z} = \text{LN}(\mathbf{Z})\mathbf{W}_k + \mathbf{b}_k, \quad (3)$$

$$\mathbf{V}_\mathbf{Z} = \text{LN}(\mathbf{Z})\mathbf{W}_v + \mathbf{b}_v, \quad (4)$$

where $\mathbf{W}_{\{q,k,v\}} \in \mathbb{R}^{D \times D}$ are the weight matrices, and $\mathbf{b}_{\{q,k,v\}} \in \mathbb{R}^D$ are the bias vectors which will be broadcast to dimensions $(N + M) \times D$ for the addition operation.

The self-attention operation for the input tokens $\mathbf{Z}$ is formulated as:

$$\mathbf{F}_\mathbf{Z} = f_{\text{SA}}(\mathbf{Z}) = \text{softmax}\left(\frac{\mathbf{Q}_\mathbf{X}\mathbf{K}_\mathbf{Z}^\top}{\sqrt{D}}\right)\mathbf{V}_\mathbf{Z}, \quad (5)$$

Note that we only need to compute the self-attention for query image tokens (i.e., $\mathbf{Q}_\mathbf{X}$), while the query prompts (i.e., $\mathbf{Q}_\mathbf{P}$) can be neglected, as shown in Fig. 1. The reason is that the output tokens computed from the query prompts will be replaced with new trainable prompts in the next ViT layer [14]. Since the MLP and its preceding LayerNorm are applied independently to each token, omitting $\mathbf{Q}_\mathbf{P}$ does not affect the output image tokens or the class token.

Equation (5) can be further expanded as Affinity, softmax and Aggregation operations:

$$\begin{cases} \mathbf{A}_\mathbf{Z} = f_{\text{aff}}(\mathbf{Q}_\mathbf{X}, \mathbf{K}_\mathbf{Z}) = \frac{\mathbf{Q}_\mathbf{X}\mathbf{K}_\mathbf{Z}^\top}{\sqrt{D}} = \frac{\mathbf{Q}_\mathbf{X}[\mathbf{K}_\mathbf{X}^\top \ \mathbf{K}_\mathbf{P}^\top]}{\sqrt{D}}, & (6a) \\ \mathbf{S}_\mathbf{Z} = \text{softmax}\left(\begin{bmatrix} \mathbf{A}_\mathbf{X} & \mathbf{A}_\mathbf{P} \end{bmatrix}\right) = \begin{bmatrix} \mathbf{S}_\mathbf{X} & \mathbf{S}_\mathbf{P} \end{bmatrix}, & (6b) \\ \mathbf{F}_\mathbf{Z} = f_{\text{agg}}(\mathbf{S}_\mathbf{Z}, \mathbf{V}_\mathbf{Z}) = \mathbf{S}_\mathbf{Z}\mathbf{V}_\mathbf{Z} = \begin{bmatrix} \mathbf{S}_\mathbf{X} & \mathbf{S}_\mathbf{P} \end{bmatrix} \begin{bmatrix} \mathbf{V}_\mathbf{X} \\ \mathbf{V}_\mathbf{P} \end{bmatrix}. & (6c) \end{cases}$$

Next, $\mathbf{F}_\mathbf{Z}$ undergoes another LayerNorm and the MLP layer to generate the output image tokens. The class token produced by the final ViT layer is then fed into the classifier for classification. Note that only the introduced prompts $\mathbf{P}$ are trainable, while the remaining pre-trained parameters in LayerNorms, qkv-transformations and MLP are kept frozen.

C. Orthogonal Projection in Convolutional Layers

A kernel in a convolutional layer can be represented as an unrolled matrix $\Theta \in \mathbb{R}^{D_{\text{in}} \times D_{\text{out}}}$, where D_{in} corresponds to the number of pixels in the convolutional kernel, and D_{out} denotes the number of kernels. Correspondingly, an input feature map can be rearranged into n_p patches, each containing D_{in} pixels. The convolution $f_{\text{conv}}(\cdot)$ is essentially the multiplication of the kernel Θ and the unrolled feature map $\mathbf{X} \in \mathbb{R}^{n_p \times D_{\text{in}}}$:

$$f_{\text{conv}}(\mathbf{X}|\Theta) = \mathbf{X}\Theta \quad (7)$$

For an input feature map $\mathbf{X}_t$ associated with task $\mathcal{T}_t$, the corresponding output feature map is expected to remain unchanged in order to overcome forgetting, even as the kernel Θ_t is updated to Θ_{t+1} in the new task $\mathcal{T}_{t+1}$. To this end, the *anti-forgetting objective* is established as:

$$f_{\text{conv}}(\mathbf{X}_t|\Theta_t) = f_{\text{conv}}(\mathbf{X}_t|\Theta_{t+1}). \quad (8)$$

We denote the kernel update in $\mathcal{T}_{t+1}$ as $\Delta\Theta$. By substituting $\Theta_{t+1} = \Theta_t + \Delta\Theta$ into (8), the above anti-forgetting objective is expanded as:

$$\mathbf{X}_t\Theta_t = \mathbf{X}_t(\Theta_t + \Delta\Theta), \quad (9)$$

which is further simplified as:

$$\mathbf{X}_t\Delta\Theta = \mathbf{0}. \quad (10)$$

We refer to (10) as the *orthogonality condition* for convolution. It suggests that during training in the new task $\mathcal{T}_{t+1}$, if the kernel update $\Delta\Theta$ can be orthogonal to the old input feature map $\mathbf{X}_t$, the corresponding output feature map will keep unchanged. In this way, the old knowledge will not be forgotten.

To implement the orthogonality condition in practice, the gradient of the kernel (denoted as $\mathbf{G}_\Theta$) is projected onto the orthogonal subspace of $\mathbf{X}_t$ during training the new task:

$$\Delta\Theta = \mathbf{B}_{\text{conv}}\mathbf{G}_\Theta, \quad (11)$$

where $\mathbf{B}_{\text{conv}}$ is an orthogonal projection matrix computed from $\mathbf{X}_t$ [10], [32], [52].

Similarly, for a ViT layer $f_{\text{vit}}(\mathbf{X}|\mathbf{P})$ with the prompts $\mathbf{P}$ to be optimized, the anti-forgetting objective is formulated as:

$f_{\text{vit}}(\mathbf{X}_t|\mathbf{P}_t) = f_{\text{vit}}(\mathbf{X}_t|\mathbf{P}_{t+1})$. However, the projection formula derived from convolution, *i.e.*, (11), cannot be applied to the ViT layer, since $f_{\text{vit}}(\mathbf{X}|\mathbf{P}) \neq \mathbf{X}\mathbf{P}$. As aforementioned, the input tokens of the ViT layer undergo the LayerNorm, three linear transformations and the self-attention. Such a complex forward propagation poses challenges in deriving orthogonal projections for prompt updates. In the next section, we will analyze the anti-forgetting objective, derive the orthogonality conditions and compute the orthogonal projection matrices for visual prompt tuning.

IV. METHOD

In this section, we begin by presenting the overall framework and derive the orthogonality conditions necessary for visual prompt tuning. We then move on to explain how orthogonal projections are computed using the null-space method. Next, we introduce the importance-aware orthogonal constraint strategy, which takes advantage of the orthogonal projection matrices derived earlier. We further extend our method to handle multi-head self-attention. Lastly, we discuss the incorporation of the classifier alignment technique.

A. Overall Framework

The framework of our approach is illustrated in Fig. 1. An image is first split into multiple patches, which are subsequently embedded as tokens. These image tokens, along with the pre-trained class token and learnable prompts, are processed through S ViT layers. The resulting output class token is fed into the classifier for classification. The cross-entropy loss $\mathcal{L}_{\text{CE}}$ is employed to train the model. Notably, only the prompts and the classifier are trainable, while the parameters of the ViT layers and the input class token are pre-trained and remain frozen.

In optimizing for a subsequent task $\mathcal{T}_{t+1}$, the gradient vectors of the prompts $\mathbf{G}_{\mathbf{P}}$ are projected onto orthogonal subspaces using particular embeddings from the previous task $\mathcal{T}_t$. These orthogonal projections apply different levels of constraint according to the dimensional importance for both the old and the new tasks. As a result, the updated prompts $\mathbf{P}_{t+1}$ preserve the output image tokens (as well as the class token) of samples from the previous task across different tasks. This method aids in preserving stable representations, thereby mitigating catastrophic forgetting. The detailed derivation of these embeddings used to form the orthogonal subspaces is provided in the following subsections.

B. Orthogonality Conditions for Prompt Tuning

We denote the input tokens for task $\mathcal{T}_t$ as $\mathbf{Z}_t = [\mathbf{X}_t; \mathbf{P}_t]$. When the prompts $\mathbf{P}_t$ are updated to $\mathbf{P}_{t+1}$ by $\Delta\mathbf{P}$ in the task $\mathcal{T}_{t+1}$, the input tokens are represented as $\mathbf{Z}_{t+1} = [\mathbf{X}_t; \mathbf{P}_{t+1}]$. We aim to analyze the following *anti-forgetting objective*:

$$f_{\text{vit}}(\mathbf{X}_t|\mathbf{P}_t) = f_{\text{vit}}(\mathbf{X}_t|\mathbf{P}_{t+1}). \quad (12)$$

Through deducing (12), we expect to derive the orthogonality conditions *represented by the prompt update $\Delta\mathbf{P}$ and related to*

the input tokens $\mathbf{X}_t$, which will serve as the basis for computing the orthogonal projection matrices.

Based on the forward propagation of the ViT layer, the anti-forgetting objective of (12) can be reduced to the following equation, which ensures that the output of the self-attention remains unchanged:

$$\mathbf{F}_{\mathbf{Z}_t} = \mathbf{F}_{\mathbf{Z}_{t+1}}. \quad (13)$$

However, when attempting to simplify the above equation based on (5), the product $\mathbf{K}_{\mathbf{Z}}^\top \mathbf{V}_{\mathbf{Z}}$ introduces a potential higher-order term $\mathbf{W}_k^\top \mathbf{Z}^\top \mathbf{Z} \mathbf{W}_v$ involving $\mathbf{Z}$. Moreover, the presence of softmax in (5) poses an additional challenge, hindering the deduction of (13).

To tackle the challenges, we propose two sufficient conditions as an alternative of (13). By deducing the two sufficient conditions, respectively, we can derive two orthogonality conditions that jointly satisfy the anti-forgetting objective. Concretely, based on the expanded form of self-attention (*i.e.*, (6)), the derivation of $\mathbf{F}_{\mathbf{Z}}$ is split into two steps: Affinity (*i.e.*, (6a)) and Aggregation (*i.e.*, (6c)). In this way, we can achieve (13) by establishing the following equations:

$$\begin{cases} f_{\text{aff}}(\mathbf{Q}_{\mathbf{X}_t}, \mathbf{K}_{\mathbf{Z}_t}) = f_{\text{aff}}(\mathbf{Q}_{\mathbf{X}_t}, \mathbf{K}_{\mathbf{Z}_{t+1}}), \\ f_{\text{agg}}(\mathbf{S}_{\mathbf{Z}_t}, \mathbf{V}_{\mathbf{Z}_t}) = f_{\text{agg}}(\mathbf{S}_{\mathbf{Z}_{t+1}}, \mathbf{V}_{\mathbf{Z}_{t+1}}). \end{cases} \quad (14a)$$

By substituting the Affinity operation in (6a) into (14a) with $\mathbf{Z}_t$ and $\mathbf{Z}_{t+1}$, respectively, and eliminating the equal terms $\sqrt{D}$ and $\mathbf{Q}_{\mathbf{X}_t} \mathbf{K}_{\mathbf{X}_t}^\top$, we derive that:

$$\mathbf{Q}_{\mathbf{X}_t} \mathbf{K}_{\mathbf{P}_t}^\top = \mathbf{Q}_{\mathbf{X}_t} \mathbf{K}_{\mathbf{P}_{t+1}}^\top, \quad (15)$$

which can be further simplified as the following equation by substituting $\mathbf{K}_{\mathbf{P}}$ using (3):

$$\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \text{LN}(\mathbf{P}_t)^\top = \mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \text{LN}(\mathbf{P}_{t+1})^\top. \quad (16)$$

With (14a) holding true, the activated attention map ((6b)) can keep unchanged, *i.e.*, $\mathbf{S}_{\mathbf{Z}_t} = \mathbf{S}_{\mathbf{Z}_{t+1}}$. Based on this derived equation and the expanded Aggregation operation in (6c), we can simplify (14b) as:

$$\mathbf{S}_{\mathbf{P}_t} \mathbf{V}_{\mathbf{P}_t} = \mathbf{S}_{\mathbf{P}_{t+1}} \mathbf{V}_{\mathbf{P}_{t+1}} = \mathbf{S}_{\mathbf{P}_t} \mathbf{V}_{\mathbf{P}_{t+1}}, \quad (17)$$

By substituting $\mathbf{V}_{\mathbf{P}}$ using (4) and omitting the constant pre-trained matrix $\mathbf{W}_v$, (17) is further simplified as:

$$\mathbf{S}_{\mathbf{P}_t} \text{LN}(\mathbf{P}_t) = \mathbf{S}_{\mathbf{P}_t} \text{LN}(\mathbf{P}_{t+1}). \quad (18)$$

Next, by substituting $\mathbf{P}_{t+1} = \mathbf{P}_t + \Delta\mathbf{P}$ into (16) and (18), we derive the following conditions for (14a) and (14b), respectively:

$$\begin{cases} \mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \text{LN}(\mathbf{P}_t)^\top = \mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \text{LN}(\mathbf{P}_t + \Delta\mathbf{P})^\top, \\ \mathbf{S}_{\mathbf{P}_t} \text{LN}(\mathbf{P}_t) = \mathbf{S}_{\mathbf{P}_t} \text{LN}(\mathbf{P}_t + \Delta\mathbf{P}). \end{cases} \quad (19a)$$

As aforementioned, we aim to deduce the orthogonality conditions expressed in terms of $\Delta\mathbf{P}$, so that we can compute orthogonal projections for the gradients of the prompts. However, the LayerNorm operation $\text{LN}(\cdot)$ causes obstacles to further simplify (19). Concretely, the square root and quadratic terms involved in computing the standard deviation σ in (1) complicate the establishment of a relationship between $\sigma_{\mathbf{P}_t + \Delta\mathbf{P}}$ and $\sigma_{\mathbf{P}_t}$ via

$\sigma_{\Delta P}$. Consequently, it becomes challenging to simplify $\text{LN}(\mathbf{P}_t)$ and $\text{LN}(\mathbf{P}_t + \Delta \mathbf{P})$ in (19).

To overcome this obstacle, we propose an invariant prompt distribution constraint. It ensures that the distribution of prompts remains unchanged during prompt updates, *i.e.*:

$$\begin{cases} \mu_{\mathbf{P}_t + \Delta \mathbf{P}} = \mu_{\mathbf{P}_t}, \\ \sigma_{\mathbf{P}_t + \Delta \mathbf{P}} = \sigma_{\mathbf{P}_t}. \end{cases} \quad (20a) \quad (20b)$$

By doing this, we can establish the relationship between $\text{LN}(\mathbf{P}_t)$ and $\text{LN}(\mathbf{P}_t + \Delta \mathbf{P})$ as:

$$\text{LN}(\mathbf{P}_t + \Delta \mathbf{P}) = \text{LN}(\mathbf{P}_t) + \frac{\Delta \mathbf{P}}{\sigma_{\mathbf{P}_t}} \odot \alpha. \quad (21)$$

By substituting (21) into (19) and eliminating the same terms $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \text{LN}(\mathbf{P}_t)^\top$ and $\mathbf{S}_{\mathbf{P}_t} \text{LN}(\mathbf{P}_t)$, we can derive that:

$$\begin{cases} \mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \Delta \mathbf{P}^\top / \sigma_{\mathbf{P}_t} \odot \alpha^\top = \mathbf{0}, \\ \mathbf{S}_{\mathbf{P}_t} \Delta \mathbf{P} / \sigma_{\mathbf{P}_t} \odot \alpha = \mathbf{0}. \end{cases} \quad (22a) \quad (22b)$$

Since α is a constant pre-trained vector and $\sigma_{\mathbf{P}_t}$ can also be treated as a constant vector under the constraint of (20), the above equation can be simplified to derive the following *orthogonality conditions* for prompt tuning:

$$\begin{cases} \mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top \Delta \mathbf{P}^\top = \mathbf{0}, \\ \mathbf{S}_{\mathbf{P}_t} \Delta \mathbf{P} = \mathbf{0}. \end{cases} \quad (23a) \quad (23b)$$

To summarize, the anti-forgetting objective in (12) can be fulfilled by achieving the two sufficient orthogonality conditions, (23a) and (23b), which are expressed in terms of $\Delta \mathbf{P}$. They reveal that if the prompts (or transposed prompts) can update in the direction orthogonal to 1) the normalized old image tokens multiplied by the qkv-transformation matrices (*i.e.*, $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top$), and 2) the activated attention map computed from old image queries and prompt keys (*i.e.*, $\mathbf{S}_{\mathbf{P}_t}$), the updated prompts will not change the outputs of previous instances and thus forgetting can be mitigated.

C. Computation of Orthogonal Projection Matrices

To satisfy the deduced orthogonality conditions, we first project the gradients of the prompts onto the subspace orthogonal to specific intermediate embeddings (*i.e.*, $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top$, $\mathbf{S}_{\mathbf{P}_t}$). The prompts are then updated using the projected gradients. To this end, we need to compute the orthogonal projection matrices that project $\mathbf{G}_P$ to $\Delta \mathbf{P}$ to satisfy (23), where $\mathbf{G}_P$ represents the gradients of the prompts.

Given that (23) defines two conditions for the single variable $\Delta \mathbf{P}$, we employ a separate optimization strategy to compute the projection matrices. Specifically, we first use a projection matrix $\mathbf{B}_1$ to project $\mathbf{G}_P^\top$ to $\Delta \mathbf{P}^\top$ to satisfy (23a): $\Delta \mathbf{P}^\top = \mathbf{B}_1 \mathbf{G}_P^\top$. Subsequently, another projection matrix $\mathbf{B}_2$ is applied to $\mathbf{G}_P$ to satisfy (23b): $\Delta \mathbf{P} = \mathbf{B}_2 \mathbf{G}_P$. By combining these two projections, the orthogonality conditions in (23) can be achieved by: $\Delta \mathbf{P} = \mathbf{B}_2 \mathbf{G}_P \mathbf{B}_1$.

In our approach, we adopt the null space method [32] to solve the two orthogonal projection matrices $\mathbf{B}_1$ and $\mathbf{B}_2$. According to [32], the bases of $\mathbf{B}_1$ and $\mathbf{B}_2$ are exactly the bases of the null spaces of $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top$ and $\mathbf{S}_{\mathbf{P}_t}$, respectively. To compute

the null-space bases of $\mathbf{B}_1$, we first obtain the uncentered covariance matrix (UCM) [32] of $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top$. Specifically, let $\mathbf{C}_1 = [\mathbf{Q}_{\mathbf{X}_t}^{(1)} \mathbf{W}_k^\top, \dots, \mathbf{Q}_{\mathbf{X}_t}^{(|\mathcal{T}_t|)} \mathbf{W}_k^\top] \in \mathbb{R}^{|\mathcal{T}_t|N \times D}$ denote the concatenated matrix of $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top$ extracted by all the $|\mathcal{T}_t|$ instances in task $\mathcal{T}_t$, where $\mathbf{Q}_{\mathbf{X}_t}^{(n)} \mathbf{W}_k^\top$ represents the n -th intermediate embedding. The UCM of $\mathbf{Q}_{\mathbf{X}_t} \mathbf{W}_k^\top$ is $\bar{\mathbf{C}}_1 = \mathbf{C}_1^\top \mathbf{C}_1 \in \mathbb{R}^{D \times D}$. Next, the singular value decomposition (SVD) is performed on the UCM $\bar{\mathbf{C}}_1$ to obtain the singular values $\Lambda_1 \in \mathbb{R}^D$ and the right singular matrix $\mathbf{U}_1 \in \mathbb{R}^{D \times D}$. After that, the right singular vectors corresponding to zero singular values are selected as the bases of null space, denoted as $\tilde{\mathbf{U}}_1 \in \mathbb{R}^{D \times R_1}$, where R_1 means the nullity. Finally, the null-space projection matrix $\mathbf{B}_1$ is calculated by: $\mathbf{B}_1 = \tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1^\top$. Similarly, $\mathbf{B}_2$ can be obtained by performing SVD on the UCM of $\mathbf{S}_{\mathbf{P}_t}$, picking the null-space bases $\tilde{\mathbf{U}}_2 \in \mathbb{R}^{M \times R_2}$, and computing the projection matrix: $\mathbf{B}_2 = \tilde{\mathbf{U}}_2 \tilde{\mathbf{U}}_2^\top \in \mathbb{R}^{M \times M}$.

Nonetheless, in practice, zero singular values may not exist due to the complexity of real-world data, making it difficult to construct the null space accurately. We propose an adaptive approach to obtain the bases of an approximated null space as a practical solution. This method selects the right singular vectors associated with singular values that are close to zero as the null-space bases. To be specific, based on the observation that the singular values in descending order form an ‘‘L’’ shape, the approximated zero singular values are defined as those below the corner point of the curve. We first obtain the index of the corner point by the maximum second derivative of the points. For example, $J_1 = \arg \max_j \{\lambda_{j-1} - 2\lambda_j + \lambda_{j+1}\}_{j=2}^{D-1}$, where λ_j is the j -th singular value. Then the right singular vectors whose corresponding singular values are less than λ_{J_1} are determined as the null-space bases to construct $\mathbf{B}_1$.

In this way, the orthogonal projections for the gradients are derived and can be expressed as:

$$\Delta \mathbf{P} = \mathbf{B}_2 \mathbf{G}_P \mathbf{B}_1 = (\tilde{\mathbf{U}}_2 \tilde{\mathbf{U}}_2^\top) \mathbf{G}_P (\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1^\top). \quad (24)$$

Additionally, there is an invariant prompt distribution constraint proposed in (20). We introduce an additional loss that minimizes the drift of the prompt distribution across tasks to achieve this constraint:

$$\mathcal{L}_{\text{LN}} = \|\mu_{\mathbf{P}_{t+1}} - \mu_{\mathbf{P}_t}\|_1 + \|\sigma_{\mathbf{P}_{t+1}} - \sigma_{\mathbf{P}_t}\|_1, \quad (25)$$

where $\mu_{\mathbf{P}_t}$ and $\sigma_{\mathbf{P}_t}$ are the fixed prompt distribution parameters serving as the targets, while $\mu_{\mathbf{P}_{t+1}}$ and $\sigma_{\mathbf{P}_{t+1}}$ are trainable distribution parameters to be optimized in the new task.

In summary, during training in the new task, we utilize (24) to perform orthogonal projections on the gradients for parameter updates. Alongside the cross-entropy loss ($\mathcal{L}_{\text{CE}}$) for classification, (25) is also incorporated to guide the training of prompts. However, enforcing a strict orthogonal constraint may restrict the plasticity of the model especially when the learning process continues over a prolonged period. To further balance the stability and plasticity, we propose an importance-aware orthogonal constraint strategy to relax the orthogonal constraints, which is elaborated in next subsection.

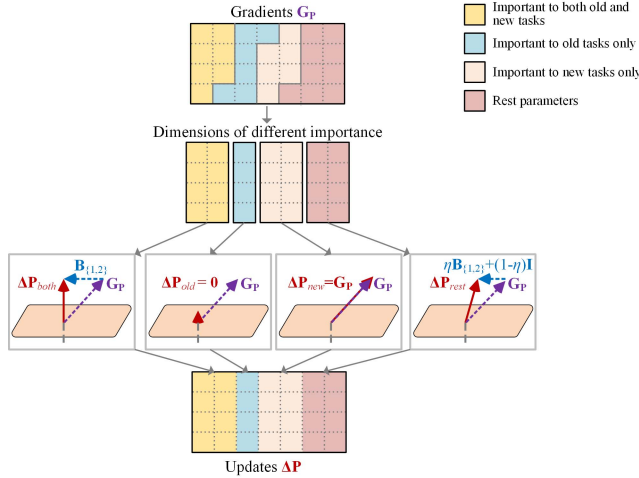


Fig. 2. Illustration of the importance-aware orthogonal constraint strategy. Prompt gradients $\mathbf{G}_P$ are partitioned into four groups at the dimension level according to parameter importance to old and new tasks. Different constraints are then applied to each group, resulting in the prompt updates $\Delta \mathbf{P}$.

D. Importance-Aware Orthogonal Constraint Strategy

Given the varying impacts of different parameters on stability and plasticity, the core idea of the importance-aware orthogonal constraint strategy is to impose distinct update constraints on prompt parameters based on their importance to old and new tasks. As illustrated in Fig. 2, we first partition the prompt parameters into four disjoint groups at the dimension level according to their importance. Parameters that are exclusively important to the old task are frozen during training on the new task, as they primarily contribute to stability. In contrast, parameters solely important to the new task are updated without orthogonal constraints, as they mainly enhance plasticity. For parameters that are important to both the old and new tasks, strict orthogonal constraints are applied to ensure updates without forgetting. Lastly, for the remaining parameters, a relaxed orthogonal constraint is employed, allowing them to acquire new knowledge while tolerating a slight sacrifice in stability.

Inspired by existing works [8], [63], [64], we utilize gradients to assess the importance of parameters to a task. At the end of training on the old task $\mathcal{T}_t$ with the dataset $\mathcal{D}_t$, the gradients of prompts are calculated through the cross-entropy loss. We denote the gradient matrix as $\mathbf{G}_{P_o} \in \mathbb{R}^{M \times D}$. After switching to the new task $\mathcal{T}_{t+1}$ and before starting training, we use a supervised contrastive loss (SCL) [65] to compute the gradients $\mathbf{G}_{P_n}$ on the dataset $\mathcal{D}_{t+1}$. The SCL is formulated as:

$$\mathcal{L}_{\text{SCL}} = \sum_{i \in \mathcal{D}_{t+1}} \frac{-1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \in \mathcal{A}(i)} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)}, \quad (26)$$

where i denotes the i -th sample in $\mathcal{D}_{t+1}$; $\mathcal{A}(i) \equiv \mathcal{D}_{t+1} \setminus \{i\}$ is the subset excluding i ; $\mathcal{P}(i) \equiv \{p \in \mathcal{A}(i) : y_p = y_i\}$ represents the subset of samples sharing the same class label as i , and $|\mathcal{P}(i)|$ represents its cardinality; $\mathbf{z}$ denotes the representation extracted by the backbone and $\tau \in \mathbb{R}^+$ is a temperature parameter. The SCL does not rely on the classifier, thus preventing inaccurate

gradients caused by the randomly initialized classifier in the new task.

Based on the obtained gradients, the corresponding importance matrices $\mathbf{E}_{P_o}$ and $\mathbf{E}_{P_n}$ are calculated by element-wise squaring [63] of the gradient matrices:

$$\mathbf{E}_{P_o} = \mathbf{G}_{P_o} \odot \mathbf{G}_{P_o}, \quad \mathbf{E}_{P_n} = \mathbf{G}_{P_n} \odot \mathbf{G}_{P_n} \in \mathbb{R}^{M \times D}. \quad (27)$$

In our approach, importance is determined at the dimension level, which means that the M elements of the vector along the prompt dimension share the same importance. The dimension-level importance is computed by summing the M importance values within the same dimension:

$$\bar{\mathbf{E}}_{P_o} = \left[\sum_{m=1}^M \mathbf{E}_{P_o}^{(m,1)}, \dots, \sum_{m=1}^M \mathbf{E}_{P_o}^{(m,D)} \right] \in \mathbb{R}^D, \quad (28)$$

where $\mathbf{E}_{P_o}^{(m,j)}$ denotes the element of the m -th row and j -th column in the matrix $\mathbf{E}_{P_o}$. Similarly, (28) can be applied to $\mathbf{E}_{P_n}$ to compute the dimensional importance vector $\bar{\mathbf{E}}_{P_n}$.

Next, we generate the initial binary masks based on the dimension-level importance vectors $\bar{\mathbf{E}}_{P_o}$ and $\bar{\mathbf{E}}_{P_n}$ to indicate the top- K important dimensions. The binary mask $\mathbf{M}_{P_o} \in \mathbb{R}^D$ is computed from $\bar{\mathbf{E}}_{P_o}$ as follows:

$$\mathbf{M}_{P_o}^{(j)} = \begin{cases} 1, & \text{if } \bar{\mathbf{E}}_{P_o}^{(j)} \text{ is in the top-}K \text{ values of } \bar{\mathbf{E}}_{P_o}, \\ 0, & \text{otherwise.} \end{cases} \quad (29)$$

The binary mask $\mathbf{M}_{P_n}$ for $\bar{\mathbf{E}}_{P_n}$ can be calculated in the same manner as in (29).

After that, we use $\mathbf{M}_{P_o}$ and $\mathbf{M}_{P_n}$ to partition the parameters into four groups, with each group represented by a binary mask. These masks indicate the dimensions important to: 1) both the old and the new tasks, 2) only the old task, 3) only the new task, and 4) the rest parameters. The corresponding binary masks, $\mathbf{M}_{both}, \mathbf{M}_{old}, \mathbf{M}_{new}, \mathbf{M}_{rest} \in \mathbb{R}^D$, are derived by:

$$\mathbf{M}_{both} = \mathbf{M}_{P_o} \odot \mathbf{M}_{P_n}, \quad (30)$$

$$\mathbf{M}_{old} = \mathbf{M}_{P_o} - \mathbf{M}_{both}, \quad (31)$$

$$\mathbf{M}_{new} = \mathbf{M}_{P_n} - \mathbf{M}_{both}, \quad (32)$$

$$\mathbf{M}_{rest} = 1 - (\mathbf{M}_{P_o} + \mathbf{M}_{P_n} - \mathbf{M}_{both}). \quad (33)$$

After obtaining the four masks, we start training the prompts in the new task $\mathcal{T}_{t+1}$ using the cross-entropy loss $\mathcal{L}_{\text{CE}}$ and the proposed invariant distribution loss $\mathcal{L}_{\text{LN}}$ as defined in (25). These masks are employed to apply different update policies to the prompt parameters during optimization:

$$\Delta \mathbf{P}_{both} = \mathbf{B}_2 \mathbf{G}_P \mathbf{B}_1 \odot \mathbf{M}_{both}, \quad (34)$$

$$\Delta \mathbf{P}_{old} = \mathbf{0} \odot \mathbf{M}_{old} = \mathbf{0}, \quad (35)$$

$$\Delta \mathbf{P}_{new} = \mathbf{G}_P \odot \mathbf{M}_{new}, \quad (36)$$

$$\Delta \mathbf{P}_{rest} = [\eta \mathbf{B}_2 + (1 - \eta) \mathbf{I}] \mathbf{G}_P [\eta \mathbf{B}_1 + (1 - \eta) \mathbf{I}] \odot \mathbf{M}_{rest}, \quad (37)$$

where $\eta \in [0, 1]$ is a hyper-parameter used to relax the orthogonality. Finally, the update formula is given by:

$$\Delta \mathbf{P} = \Delta \mathbf{P}_{both} + \Delta \mathbf{P}_{old} + \Delta \mathbf{P}_{new} + \Delta \mathbf{P}_{rest}. \quad (38)$$

Comparing with (24), the proposed importance-aware orthogonal constraint strategy applies strict orthogonal constraints only to parameters significant for both the old and new tasks (34). Parameters crucial to the old task are frozen (35), while those essential to the new task are updated along their original direction (36). The remaining parameters, which moderately contribute to both old and new tasks, are updated with relaxed orthogonal constraints (37). This strategy offers a flexible and effective approach to achieve a balance between stability and plasticity.

E. Extension to Multi-Head Attention

As multi-head self-attention is widely utilized in prevalent attention-based architectures, we extend the orthogonality conditions in (23) from a single head to multiple heads. For the h -th head, the intermediate embeddings in (23) are denoted as $\mathbf{Q}_{\mathbf{x}_{t,h}} \in \mathbb{R}^{N \times d}$, $\mathbf{W}_{k,h} \in \mathbb{R}^{D \times d}$ and $\mathbf{S}_{\mathbf{p}_{t,h}} \in \mathbb{R}^{N \times M}$, respectively, where $d = D/H$ represents the token dimension in each head, and H is the total number of heads. To satisfy the anti-forgetting objective, the orthogonality conditions should be satisfied in each head, *i.e.*:

$$\forall h \in \{1, 2, \dots, H\}, \begin{cases} \mathbf{Q}_{\mathbf{x}_{t,h}} \mathbf{W}_{k,h}^\top \Delta \mathbf{P}^\top = \mathbf{0}, & (39a) \\ \mathbf{S}_{\mathbf{p}_{t,h}} \Delta \mathbf{P} = \mathbf{0}. & (39b) \end{cases}$$

Although (39a) contains a total of H conditions, they can be unified into a single equation. Concretely, we define $\Omega_{1,t} = [\mathbf{Q}_{\mathbf{x}_{t,1}} \mathbf{W}_{k,1}^\top; \dots; \mathbf{Q}_{\mathbf{x}_{t,H}} \mathbf{W}_{k,H}^\top] \in \mathbb{R}^{HN \times D}$ to denote the concatenated intermediate embeddings $\mathbf{Q}_{\mathbf{x}_{t,h}} \mathbf{W}_{k,h}^\top$ from all heads. By leveraging the properties of block matrix multiplication, (39a) can be formulated as: $\Omega_{1,t} \Delta \mathbf{P}^\top = \mathbf{0}$. Similarly, (39b) is reformulated as $\Omega_{2,t} \Delta \mathbf{P} = \mathbf{0}$, where $\Omega_{2,t} = [\mathbf{S}_{\mathbf{p}_{t,1}}; \dots; \mathbf{S}_{\mathbf{p}_{t,H}}] \in \mathbb{R}^{HN \times M}$ denotes the concatenation of $\mathbf{S}_{\mathbf{p}_{t,h}}$ across all heads.

In summary, the orthogonality conditions for multi-head self-attention are expressed as:

$$\begin{cases} \Omega_{1,t} \Delta \mathbf{P}^\top = \mathbf{0}, & (40a) \\ \Omega_{2,t} \Delta \mathbf{P} = \mathbf{0}. & (40b) \end{cases}$$

They indicate that the primary difference between single-head and multi-head orthogonality conditions lies in the subspace. For multi-head attention, the subspace to which the prompt updates should be orthogonal is spanned by the concatenated intermediate embeddings from all heads. Note that the computation of orthogonal projection matrices, the invariant prompt distribution loss and the importance-aware orthogonal constraint strategy are still applicable to the multi-head variant.

F. Classifier Alignment on No-drifting Representations

When evaluating on data from previous tasks, the highest score among classifiers from other tasks may exceed that of the classifier trained on the corresponding task. This issue, referred to as the inter-task confusion [48], [50], arises from the independent training of classifiers for each task. Several works [22], [42], [66] attempt to mitigate this problem by recording the representation distributions of old data, sampling

from these recorded distributions and jointly fine-tuning all available classifiers (*i.e.*, classifier alignment). However, since the representations of old data will drift (*i.e.*, change) with model updates, the recorded distributions may become invalid for training classifiers. Existing methods either disregard the representation drift [22], [42] or rely on the semantic drift compensation [66], [67]. In contrast, our approach can seamlessly combine with the classifier alignment to maximize its effectiveness in addressing inter-task confusion. This is attributed to its ability to maintain the representations (almost) unchanged, ensuring that the recorded distributions remain consistently valid for aligning classifiers.

Specifically, after training each task, we extract the representations from the backbone for each class within the task. The set of representations for the c -th class is denoted as $\varphi_c = \{\mathbf{z}_c^{(1)}, \dots, \mathbf{z}_c^{(n_c)}\}$, where $\mathbf{z}_c^{(j)} \in \mathbb{R}^D$ is the representation of the j -th sample, and n_c is the number of samples belonging to this class. The mean vector $\boldsymbol{\mu}_c \in \mathbb{R}^D$ and the covariance matrix $\boldsymbol{\Sigma}_c \in \mathbb{R}^{D \times D}$ of φ_c are then computed and recorded. After that, a number of representations are sampled from the Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$, where c spans all the classes learned so far. The sampled representations are used to jointly fine-tune the classifiers of all known tasks. Finally, during evaluation, we use the pre-trained encoder with optimized prompts and the fine-tuned classifiers for classification. An algorithm for our full approach is provided in Algorithm 1.

V. EXPERIMENTS

A. Experimental Setups

1) *Benchmarks*: We validate our approach under three continual learning (CL) scenarios: conventional CL, long-sequence CL and out-of-distribution (OOD) CL, utilizing a total of five datasets in our experiments.

The conventional CL and long-sequence CL scenarios involve the ImageNet-R [68], CIFAR-100 [69], DomainNet [70] and OmniBenchmark [71] datasets. ImageNet-R consists of 200 classes with diverse styles, such as paintings, embroidery, and cartoons. Following [18], [20], we randomly split these classes into 10 or 20 tasks (referred to as 10/20-split ImageNet-R) with an equal number of classes per task, which forms two class-incremental learning (CIL) benchmarks under the conventional CL scenario. CIFAR-100 is a classical natural image classification dataset with 100 classes, which are split into 10 CIL tasks. The original DomainNet was a multi-domain classification dataset comprising six domains (*e.g.*, paintings, sketches, cliparts) and a total of 345 categories. We follow [25], [48] to select the top 200 classes with the largest number of images to construct a cross-domain CIL dataset, with 20 classes per CIL task in the conventional CL setting. Following Zhou et al. [72], we adopt OmniBenchmark as a large-scale CIL benchmark to demonstrate the scalability of our method. OmniBenchmark contains 95,682 images across 300 classes, which is three times larger than ImageNet-R and includes 100 more classes. For consistency with [72], we follow the conventional CL scenario by splitting OmniBenchmark into 10 tasks. In the long-sequence

Algorithm 1: Algorithm of the proposed method VPT-NSP²⁺⁺.

Require: Datasets $\mathcal{D}_t = \{(x_t^{(n)}, y_t^{(n)})\}_{n=1}^{|\mathcal{T}_t|}$ for task $\mathcal{T}_t \in \{\mathcal{T}_1, \mathcal{T}_2, \dots\}$, ViT model $f_{\text{model}}(\cdot | \mathbf{P}_t)$ with the prompts $\mathbf{P}_t$ to be optimized, uncentered covariance matrices $\bar{\mathbf{C}}_1$ and $\bar{\mathbf{C}}_2$, projection matrices $\mathbf{B}_1$ and $\mathbf{B}_2$, masks $\mathbf{M}_{\{\text{both}, \text{old}, \text{new}, \text{rest}\}}$, learning rate γ

Ensure: Optimized prompts $\mathbf{P}_t$

- 1: **Initialization:** Randomly initialize $\mathbf{P}_t$; $\bar{\mathbf{C}}_1 = \mathbf{0}$, $\bar{\mathbf{C}}_2 = \mathbf{0}$, $\mathbf{B}_1 = \mathbf{I}$, $\mathbf{B}_2 = \mathbf{I}$
- 2: **for** task $\mathcal{T}_t \in \{\mathcal{T}_1, \mathcal{T}_2, \dots\}$ **do**
- 3: */* Compute importance-aware binary masks */*
- 4: **if** $t > 1$ **then**
- 5: Obtain gradients $\mathbf{G}_{\mathbf{P}_n}$ by back propagation of $\mathcal{L}_{\text{SCL}}$
- 6: Get masks $\mathbf{M}_{\{\text{both}, \text{old}, \text{new}, \text{rest}\}}$ by (27)~(33)
- 7: **end if**
- 8: */* Optimize prompts */*
- 9: **repeat**
- 10: Sample a mini-batch $\mathbf{x}_t, \mathbf{y}_t \sim \mathcal{D}_t$
- 11: Obtain prediction by $\hat{\mathbf{y}}_t \leftarrow f_{\text{model}}(\mathbf{x}_t | \mathbf{P}_t)$
- 12: Compute cross-entropy loss $\mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}_t, \mathbf{y}_t)$
- 13: **if** $t > 1$ **then**
- 14: Compute $\mathcal{L}_{\text{LN}}$ by (25)
- 15: Calculate the total loss $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{LN}}$
- 16: Compute gradients $\mathbf{G}_{\mathbf{P}}$ using the total loss $\mathcal{L}_{\text{total}}$
- 17: Compute $\Delta \mathbf{P}$ by (34)~(38)
- 18: **else**
- 19: Compute gradients $\mathbf{G}_{\mathbf{P}}$ using the loss $\mathcal{L}_{\text{CE}}$
- 20: $\Delta \mathbf{P} \leftarrow \mathbf{G}_{\mathbf{P}}$
- 21: **end if**
- 22: Update the prompts by $\mathbf{P}_t \leftarrow \mathbf{P}_t - \gamma \times \Delta \mathbf{P}$
- 23: **until** convergence
- 24: */* Compute null-space projection matrices */*
- 25: Initialize two temporary matrices $\mathbf{C}_1 = [\]$ and $\mathbf{C}_2 = [\]$
- 26: **for** $x_t^{(n)} \in \mathcal{D}_t$ **do**
- 27: Get $\mathbf{Q}_{\mathbf{x}_t}^{(n)} \mathbf{W}_k^\top$ and $\mathbf{S}_{\mathbf{P}_t}^{(n)}$ by $f_{\text{model}}(x_t^{(n)} | \mathbf{P}_t)$
- 28: Update $\mathbf{C}_1$ and $\mathbf{C}_2$ by concatenating $\mathbf{Q}_{\mathbf{x}_t}^{(n)} \mathbf{W}_k^\top$ and $\mathbf{S}_{\mathbf{P}_t}^{(n)}$, respectively
- 29: **end for**
- 30: $\bar{\mathbf{C}}_1 \leftarrow \bar{\mathbf{C}}_1 + \mathbf{C}_1^\top \mathbf{C}_1$, $\bar{\mathbf{C}}_2 \leftarrow \bar{\mathbf{C}}_2 + \mathbf{C}_2^\top \mathbf{C}_2$
- 31: Get null-space bases $\tilde{\mathbf{U}}_1, \tilde{\mathbf{U}}_2$ by SVD on $\bar{\mathbf{C}}_1, \bar{\mathbf{C}}_2$
- 32: $\mathbf{B}_1 \leftarrow \tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1^\top$, $\mathbf{B}_2 \leftarrow \tilde{\mathbf{U}}_2 \tilde{\mathbf{U}}_2^\top$
- 33: Get gradients $\mathbf{G}_{\mathbf{P}_o}$ by back propagation of $\mathcal{L}_{\text{CE}}$
- 34: Classifier alignment as introduced in Section IV-F
- 35: **end for**

CL scenario, ImageNet-R and DomainNet are split into 50 and 100 CIL tasks, and CIFAR-100 is split into 50 tasks.

The OOD CL scenario means that the domain of downstream tasks is extremely different from the pre-training data, *e.g.*, images captured by specialized equipment. We adopt two OOD

datasets in our experiments: EuroSAT [73] and RESISC45 [74]. EuroSAT contains 10 classes of satellite images for land use and land cover classification, such as forest, highway and residential. They are split into 2 and 5 tasks with disjoint classes to form CIL benchmarks. The RESISC dataset has 45 scene classes (*e.g.*, harbor, airport, thermal power station) for remote sensing image scene classification. We split these classes into 5 and 9 tasks for continual learning.

2) *Backbone:* Following previous works, we use the ViT-B/16 [111] pre-trained on ImageNet-21k [75] as the backbone, and adopt VPT [14] to continually fine-tune our model on sequential downstream tasks. In the ViT-B/16, the token dimension (D) is 768. The prompt length (M) is set to 4 by default, and prompts are inserted into all of the 12 ViT layers. The sequentially fine-tuned baseline without anti-forgetting techniques is denoted as “VPT-Seq”, while the approach proposed in this work is denoted as “VPT-NSP²⁺⁺” to differentiate it from our prior work “VPT-NSP²” [33].

3) *Implementations Details:* Training images are augmented using AutoAugment [76] and resized to 224×224 pixels before being input into the model. We use the Adam optimizer [77] ($\beta_1=0.9$, $\beta_2=0.999$) to train the model for 100 epochs with a batch size of 240. The learning rate is adjusted from 0.01 to 1×10^{-3} and 1×10^{-4} at the 50-th and 80-th epochs, respectively. The training loss is comprised of the cross-entropy loss ($\mathcal{L}_{\text{CE}}$) and the invariant prompt distribution loss ($\mathcal{L}_{\text{LN}}$, (25)), with their coefficients both set to 1. We set the temperature in the cross-entropy loss to 25 for CIFAR-100 and 30 for other datasets. The temperature τ in the SCL loss ($\mathcal{L}_{\text{SCL}}$, (26)) is set to 1. In (29), the top-50 important dimensions ($K=50$) are selected to generate the initial binary masks. The hyper-parameter η used for relaxing orthogonal constraints in (37) is set to 0.96 by default. Both hyper-parameters K and η remain fixed across all experiments, covering all tasks and datasets. During the classifier alignment, all classifiers are jointly trained by the SGD optimizer for 20 epochs, with a learning rate of 1×10^{-3} , momentum of 0.9, and weight decay of 1×10^{-4} .

We conduct experiments using four NVIDIA GeForce RTX 4090 GPUs on a server with 128 GB of RAM. Our code is implemented in PyTorch [78] with the support of the ‘timm’ library [79]. The code is available at: <https://github.com/zugexiaodui/VPT-NSP2pp>.

4) *Metrics:* Following L2P [17] and subsequent prompt-tuning-based works, performance is primarily measured by the final average accuracy, which is defined as: $\text{Accuracy} = \frac{1}{T} \sum_{i=1}^T a_{T,i}$, where $a_{T,i}$ represents the accuracy of the model evaluated on the i -th task samples after completing training on the T -th task. Additionally, we report the final average forgetting, defined as: $\text{Forgetting} = \frac{1}{T-1} \sum_{i=1}^{T-1} \max_{j \in \{1, 2, \dots, T-1\}} (a_{j,i} - a_{T,i})$. Higher accuracy indicates better performance, while lower forgetting reflects stronger stability. Note that accuracy serves as the primary metric of interest, as it captures both stability and plasticity and directly reflects the practical classification performance. We report the mean results over three runs in our experiments.

TABLE I

COMPARISON WITH EXISTING METHODS UNDER THE CONVENTIONAL CL SETTING. THE STANDARD DEVIATION (I.E., THE VALUE AFTER $\pm$) IS ALSO REPORTED WHEN AVAILABLE. MISSING RESULTS IN THE CORRESPONDING PAPERS ARE INDICATED WITH “-”. ACCURACIES MARKED WITH $\dagger$, $\ddagger$ AND * ARE REPRODUCED BY [25], [72] AND US, RESPECTIVELY, DUE TO THE LACK OF OFFICIAL RESULTS. THE BEST ACCURACY AMONG ALL METHODS IS HIGHLIGHTED IN BOLD, WHILE THE BEST ACCURACY AMONG EXISTING METHODS (EXCLUDING OUR WORKS) IS UNDERLINED.

Method	Venue	10S-ImageNet-R		20S-ImageNet-R		10S-CIFAR-100		10S-DomainNet		10S-OmniBench	
		Acc. $\uparrow$	Forgetting $\downarrow$	Acc. $\uparrow$	Forgetting $\downarrow$	Acc. $\uparrow$	Forgetting $\downarrow$	Acc. $\uparrow$	Forgetting $\downarrow$	Acc. $\uparrow$	Forgetting $\downarrow$
L2P [17]	CVPR’22	61.57 $\pm$ 0.66	9.73 $\pm$ 0.47	59.38 $\pm$ 0.50	5.89 $\pm$ 0.36	83.83 $\pm$ 0.04	7.63 $\pm$ 0.30	81.17 $\pm$ 0.83 $\dagger$	8.98 $\pm$ 1.25	64.69 $\ddagger$	-
DualPrompt [18]	ECCV’22	68.13 $\pm$ 0.49	4.68 $\pm$ 0.20	63.21 $\pm$ 0.49	5.28 $\pm$ 0.45	86.51 $\pm$ 0.33	5.16 $\pm$ 0.09	81.70 $\pm$ 0.78 $\dagger$	8.04 $\pm$ 0.31	65.52 $\ddagger$	-
CODA-Prompt [20]	CVPR’23	75.45 $\pm$ 0.56	1.64 $\pm$ 0.10	72.37 $\pm$ 1.19	0.96 $\pm$ 0.15	86.25 $\pm$ 0.74	1.67 $\pm$ 0.26	80.04 $\pm$ 0.79 $\dagger$	10.16 $\pm$ 0.35	68.09 $\ddagger$	-
APG [22]	ICCV’23	73.27	8.59	71.22	7.39	89.35	6.01	-	-	-	-
LAE [41]	ICCV’23	72.66 $\pm$ 0.63	-	69.67 $\pm$ 0.86	-	85.59 $\pm$ 0.46	-	-	-	70.63 $\ddagger$	-
LGCL [21]	ICCV’23	69.46 $\pm$ 0.04	4.20 $\pm$ 0.06	-	-	87.23 $\pm$ 0.21	5.10 $\pm$ 0.15	-	-	-	-
C-LN [80]	ICCVW’23	76.36 $\pm$ 0.51	8.31 $\pm$ 1.28	71.72 $\pm$ 0.47	5.42 $\pm$ 0.28	86.95 $\pm$ 0.37	6.98 $\pm$ 0.43	-	-	-	-
ESN [48]	AAAI’23	71.07 $\pm$ 0.29	4.99 $\pm$ 0.49	64.77 $\pm$ 0.71	6.65 $\pm$ 1.24	86.34 $\pm$ 0.52	4.76 $\pm$ 0.14	79.22 $\pm$ 2.04 $\dagger$	10.62 $\pm$ 2.12	-	-
C-ADA [81]	ECCV’24	76.66	-	73.47	-	87.18	-	-	-	-	-
OS-Prompt++ [82]	ECCV’24	75.67 $\pm$ 0.40	1.27 $\pm$ 0.10	73.77 $\pm$ 0.19	0.79 $\pm$ 0.07	86.68 $\pm$ 0.67	1.18 $\pm$ 0.21	-	-	-	-
HSICBO [83]	AAAI’24	71.43 $\pm$ 0.22	-	-	-	-	-	-	-	-	-
EvoPrompt [27]	AAAI’24	76.83 $\pm$ 0.08	2.78 $\pm$ 0.06	74.41 $\pm$ 0.23	2.56 $\pm$ 0.22	87.97 $\pm$ 0.30	2.60 $\pm$ 0.42	79.50 $\pm$ 0.29*	3.81 $\pm$ 0.36	-	-
PGP [84]	ICLR’24	69.34 $\pm$ 0.05	4.53 $\pm$ 0.04	-	-	86.92 $\pm$ 0.05	5.35 $\pm$ 0.19	80.41 $\pm$ 0.25*	8.39 $\pm$ 0.18	-	-
OVOR-Deep [50]	ICLR’24	76.11 $\pm$ 0.21	7.16 $\pm$ 0.34	73.85 $\pm$ 0.29	6.80 $\pm$ 0.65	85.99 $\pm$ 0.89	6.42 $\pm$ 2.03	79.61 $\pm$ 0.86*	4.77 $\pm$ 0.94	-	-
ConvPrompt [29]	CVPR’24	77.86 $\pm$ 0.25	4.33 $\pm$ 0.24	75.10 $\pm$ 0.39	4.10 $\pm$ 0.29	88.87 $\pm$ 0.33	4.75 $\pm$ 0.15	79.47 $\pm$ 0.35*	6.49 $\pm$ 0.43	-	-
InfLoRA [44]	CVPR’24	75.65 $\pm$ 0.14	5.73 $\pm$ 0.44	71.01 $\pm$ 0.45	6.83 $\pm$ 0.44	87.06 $\pm$ 0.25	6.22 $\pm$ 0.39	81.45 $\pm$ 0.68*	5.35 $\pm$ 0.52	72.69 $\pm$ 0.50*	10.54 $\pm$ 0.63
EASE [45]	CVPR’24	76.17	7.82	73.27	8.51	87.76	5.94	78.89*	7.89	70.30 $\pm$ 0.44*	11.79 $\pm$ 0.42
CPrompt [25]	CVPR’24	77.14 $\pm$ 0.11	5.97 $\pm$ 0.68	74.79 $\pm$ 0.28*	7.34 $\pm$ 0.65	87.82 $\pm$ 0.21	5.06 $\pm$ 0.50	82.97 $\pm$ 0.34	7.45 $\pm$ 0.93	74.25 $\pm$ 0.58*	10.33 $\pm$ 0.60
Aper [72]	IJCV’25	74.53 $\pm$ 0.42*	7.66 $\pm$ 0.55	71.22 $\pm$ 0.51*	8.96 $\pm$ 0.63	87.85 $\pm$ 0.38*	4.85 $\pm$ 0.31	82.63 $\pm$ 0.69*	6.68 $\pm$ 0.58	74.35 $\pm$ 0.40*	8.77 $\pm$ 0.45
VPT-Seq	Baseline	72.46 $\pm$ 0.54	19.41 $\pm$ 0.33	66.95 $\pm$ 0.65	25.89 $\pm$ 0.52	87.27 $\pm$ 0.68	12.33 $\pm$ 0.55	73.28 $\pm$ 0.82	25.65 $\pm$ 0.92	62.59 $\pm$ 0.65	22.37 $\pm$ 0.71
VPT-NSP ² [33]	Our prior work	78.88 $\pm$ 0.50	5.06 $\pm$ 0.26	76.13 $\pm$ 0.57	5.10 $\pm$ 0.37	91.74 $\pm$ 0.63	3.28 $\pm$ 0.45	83.54 $\pm$ 0.77	8.54 $\pm$ 0.48	74.77 $\pm$ 0.52	9.25 $\pm$ 0.70
VPT-NSP ²⁺⁺	This work	80.06 $\pm$ 0.48	4.70 $\pm$ 0.22	77.62 $\pm$ 0.51	5.17 $\pm$ 0.25	92.15 $\pm$ 0.55	3.17 $\pm$ 0.31	85.18 $\pm$ 0.54	5.82 $\pm$ 0.32	76.22 $\pm$ 0.51	8.53 $\pm$ 0.74
Joint training	Upper-bound	84.60 $\pm$ 0.24	-	84.60 $\pm$ 0.24	-	93.87 $\pm$ 0.25	-	89.25 $\pm$ 0.32	-	80.47 $\pm$ 0.27	-

B. Comparison with Existing Methods

Table I presents the comparison of the proposed approach with existing methods and the baseline under the conventional CL scenario. The key conclusion can be obtained as follows: (1) The proposed VPT-NSP²⁺⁺ demonstrates significant effectiveness in mitigating forgetting. Compared to the baseline VPT-Seq, it improves accuracy by 4.9%~13.6% (with an average increase of 9.74%) and reduces forgetting by 9.2%~20.7% (with an average decrease of 15.7%). (2) Our improved approach VPT-NSP²⁺⁺ proposed in this work outperforms our prior work VPT-NSP² [33]. Specifically, the accuracy improves by an average of 1.23% and the forgetting declines by an average of 0.78%. (3) VPT-NSP²⁺⁺ achieves state-of-the-art performance on the five conventional CL benchmarks, surpassing 11 newly published approaches in 2024 and 2025. It exceeds the highest accuracy among these methods by an average margin of 2.5%. Even on the large-scale OmniBenchmark, our method achieves 1.9% higher accuracy and lower forgetting than the state-of-the-art method Aper, demonstrating the strong effectiveness and scalability of our method.

C. Generalizability Validation

We validate the generalizability of our approach from three perspectives: using different pre-training parameters, under the long-sequence CL scenario, and under the OOD CL scenario.

1) *Different pre-training datasets and architectures*: As mentioned before, the ViT pre-trained on ImageNet-21k is used in our default setting. In this experiment, we employ four additional types of ViT parameters pre-trained using various

paradigms and datasets. They are obtained by: naive classification on ImageNet-1k, DINO [85] (self-supervised pre-training) on ImageNet-1k, MILL (semantic softmax pre-training) on ImageNet-21k-P [86] and CLIP [87] (contrastive language-image pre-training) on LAION-2B [88] (only the image encoder of CLIP is utilized). The performance of VPT-NSP²⁺⁺ compared to the baseline VPT-Seq is shown in Fig. 3. Our approach consistently demonstrates significant anti-forgetting capabilities, regardless of the pre-training paradigm (e.g., supervised or self-supervised learning) or dataset type (e.g., images or image-text pairs).

We further conduct experiments on various ViT architectures, including ViT-Base/32, ViT-Small/16, ViT-Small/32, and ViT-Large/16, which are all pre-trained on ImageNet-21k. As shown in Fig. 4, our method consistently improves the accuracy by 12.4% on average across all architectures while reducing the forgetting by 22.8%. These results highlight the robustness and generalizability of our method across diverse pre-training settings.

2) *Long-sequence continual learning*: To validate the effectiveness of our approach in addressing the long-sequence CL problem, we conduct experiments on five benchmarks with task counts of 50 and 100. In addition, we reproduce six existing approaches, as shown in Table II. Our proposed approach VPT-NSP²⁺⁺ improves accuracy by an average of 17.5% compared to the baseline VPT-Seq, and by 5.1% compared to our prior work VPT-NSP². Moreover, it exceeds the best existing approach(es) by 4.5~9.7%, with an average accuracy improvement of 7.1%. Long-term CL requires a more effective balance between stability and plasticity. Compared to the conventional CL scenario, the relative advantages of VPT-NSP²⁺⁺ are further amplified

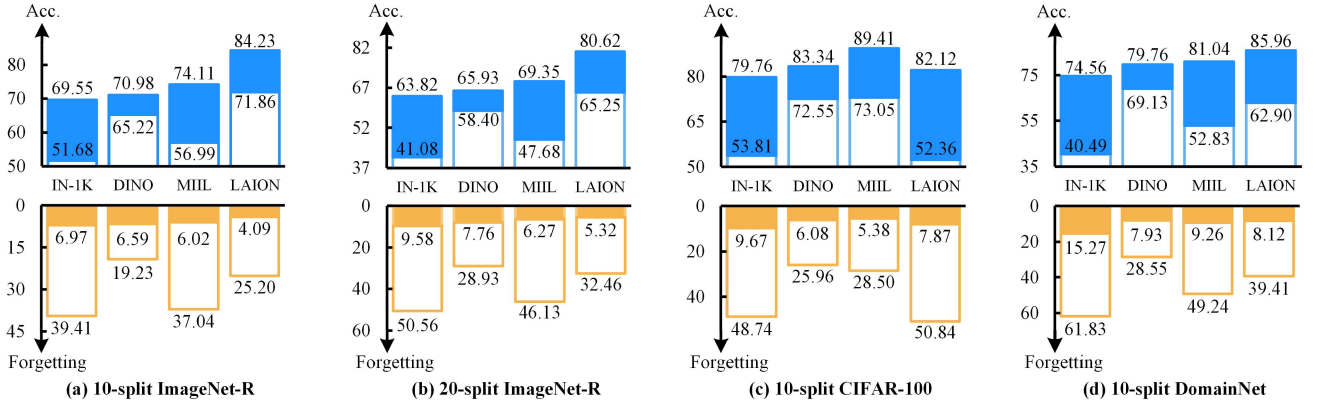


Fig. 3. Accuracy and forgetting bars of using different types of pre-training parameters. The values annotated on filled bars denote the accuracy or forgetting of VPT-NSP²⁺⁺, while the values of blank bars represent the two metrics of VPT-Seq. “IN-1k”: the pre-training parameters are trained by naive classification on ImageNet-1k; “DINO”: DINO on ImageNet-1k; “MIIL”: MIIL on ImageNet-21k-P; “LAION”: CLIP on LAION-2B.

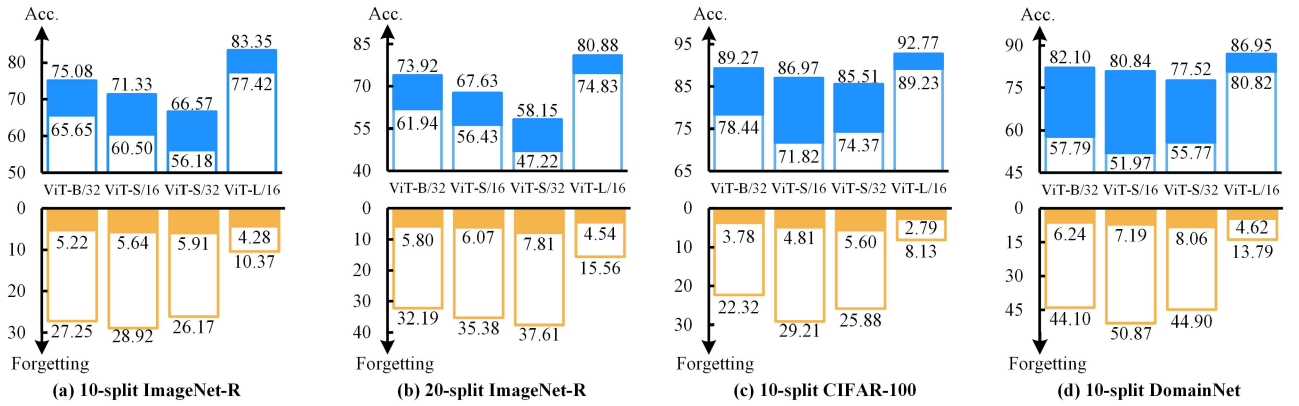


Fig. 4. Experiments on ViTs with diverse architectures (ViT-Base/32, ViT-Small/16, ViT-Small/32, and ViT-Large/16) pre-trained on ImageNet-21k. The values annotated on filled bars denote the accuracy or forgetting of VPT-NSP²⁺⁺, while the values of blank bars represent the two metrics of VPT-Seq.

TABLE II
RESULTS ACROSS FIVE BENCHMARKS UNDER THE LONG-SEQUENCE CONTINUAL LEARNING SCENARIO

Method	Venue	50S-ImageNet-R		50S-CIFAR-100		50S-DomainNet		100S-ImageNet-R		100S-DomainNet	
		Acc.↑	Forgetting↓	Acc.↑	Forgetting↓	Acc.↑	Forgetting↓	Acc.↑	Forgetting↓	Acc.↑	Forgetting↓
L2P [17]	CVPR'22	48.53±0.51	12.99±0.36	76.19±0.72	12.06±0.57	59.45±0.75	11.53±0.60	38.87±0.65	15.26±0.64	50.52±0.44	17.66±0.67
EvoPrompt [27]	AAAI'24	68.53±0.66	10.03±0.45	76.60±0.87	13.86±0.53	67.68±0.77	10.41±0.68	61.84±0.71	15.84±0.74	56.35±0.58	21.39±0.76
OVOR-Deep [50]	ICLR'24	60.08±0.68	5.86±0.60	65.69±0.74	14.28±0.63	66.27±0.71	7.43±0.61	40.49±0.80	8.12±0.66	47.65±0.78	8.91±0.79
InfLoRA [44]	CVPR'24	59.02±0.47	11.02±0.32	61.49±0.55	13.68±0.36	69.96±0.46	9.51±0.55	38.16±0.51	15.11±0.49	44.32±0.72	17.85±0.60
EASE [45]	CVPR'24	68.17±0.79	7.76±0.90	74.47±0.86	9.31±0.91	61.20±0.87	10.01±0.84	47.55±0.87	8.22±0.85	33.08±0.89	32.14±0.91
CPrompt [25]	CVPR'24	68.47±0.67	8.16±0.57	74.97±0.61	7.45±0.50	67.87±0.56	9.36±0.68	56.95±0.69	10.20±0.72	53.73±0.62	12.14±0.66
VPT-Seq	Baseline	56.38±0.85	37.91±0.94	70.47±0.97	29.21±0.90	58.39±0.92	44.79±0.88	49.72±0.91	45.53±0.95	46.39±0.93	49.34±1.02
VPN-NSP ² [33]	Our prior work	69.48±0.49	6.51±0.64	82.98±0.67	6.66±0.79	71.28±0.57	11.36±0.55	62.23±0.62	12.13±0.65	57.35±0.72	13.82±0.68
VPT-NSP ²⁺⁺	This work	74.91 ±0.58	6.99 ±0.53	86.27 ±0.73	7.33 ±0.59	75.86 ±0.62	8.39 ±0.61	66.33 ±0.71	8.17 ±0.73	65.33 ±0.77	9.30 ±0.75
Joint training	Upper-bound	84.60±0.24	-	93.87±0.25	-	89.25±0.32	-	84.60±0.24	-	89.25±0.32	-

in the long-sequence setting, demonstrating its superior capability in preventing catastrophic forgetting and adapting to new tasks.

3) *Out-of-distribution continual learning*: In real-world deployment, models inevitably encounter new data that do not align with the distribution of pre-training data. We evaluate our approach on four OOD benchmarks using the EuroSAT and

RESISC45 datasets. The results are presented in Table III. Compared to the baseline, the accuracy of VPT-NSP²⁺⁺ increases by 26.5% and the forgetting decreases by 42.7% in average. Moreover, VPT-NSP²⁺⁺ surpasses our previous work VPT-NSP² by an average accuracy gain of 4.6%. When compared to existing methods, VPT-NSP²⁺⁺ outperforms the highest accuracies among them by an average margin of 2.84%.

TABLE III
RESULTS ACROSS FOUR BENCHMARKS UNDER THE OUT-OF-DISTRIBUTION CONTINUAL LEARNING SCENARIO

Method	Venue	2S-EuroSAT		5S-EuroSAT		5S-RESISC		9S-RESISC	
		Acc.↑	Forgetting↓	Acc.↑	Forgetting↓	Acc.↑	Forgetting↓	Acc.↑	Forgetting↓
L2P [17]	CVPR'22	89.27±0.30	16.53±0.35	50.13±0.41	47.45±0.45	76.72±0.46	15.31±0.40	55.54±0.58	27.57±0.48
EvoPrompt [27]	AAAI'24	95.70±0.35	5.72±0.42	76.43±0.42	14.59±0.49	84.89±0.47	11.22±0.52	76.75±0.63	15.45±0.51
OVOR-Deep [50]	ICLR'24	94.48±0.33	3.56±0.37	84.35±0.39	8.94±0.55	85.83±0.36	9.40±0.44	80.52±0.55	9.65±0.69
InfLoRA [44]	CVPR'24	96.59±0.27	2.71±0.40	81.26±0.41	11.13±0.53	86.17±0.43	9.32±0.45	79.71±0.56	12.09±0.64
EASE [45]	CVPR'24	94.22±0.39	8.24±0.49	79.96±0.47	11.92±0.55	81.68±0.51	14.43±0.54	75.97±0.66	17.73±0.73
CPrompt [25]	CVPR'24	93.33±0.36	8.68±0.38	79.80±0.35	7.05±0.43	86.29±0.40	7.20±0.45	79.98±0.45	8.72±0.57
VPT-Seq	Baseline	65.89±0.45	62.24±0.64	63.96±0.52	42.42±0.59	72.38±0.62	32.66±0.65	50.97±0.79	54.09±0.80
VPN-NSP ² [33]	Our prior work	95.04±0.26	1.59±0.33	80.67±0.34	5.81±0.53	86.79±0.47	5.16±0.53	78.32±0.59	8.97±0.72
VPT-NSP ² ++	This work	97.37 ±0.34	1.83±0.44	89.77 ±0.39	5.85±0.58	89.21 ±0.46	4.52±0.48	82.75 ±0.60	8.27±0.65
Joint training	Upper-bound	98.93±0.32	-	98.93±0.32	-	96.35±0.29	-	96.35±0.29	-

TABLE IV
EFFECTS OF DIFFERENT PROPOSED COMPONENTS. “Ms” REPRESENTS THE FOUR BINARY MASKS OF THE IMPORTANCE-AWARE ORTHOGONAL CONSTRAINT STRATEGY. “CA” STANDS FOR CLASSIFIER ALIGNMENT. THE FIRST ROW IN THE FIRST SECTION IS VPT-SEQ, THE LAST ROW IN THE SECOND SECTION REPRESENTS VPT-NSP², AND THE LAST ROW IN THE THIRD SECTION CORRESPONDS TO THE PROPOSED VPT-NSP²++.

B ₁	B ₂	L _{LN}	Ms	CA	10S-IN-R		10S-Domain		50S-IN-R		9S-RESISC	
					Acc.↑	Fgt.↓	Acc.↑	Fgt.↓	Acc.↑	Fgt.↓	Acc.↑	Fgt.↓
					72.46	19.41	73.28	25.65	56.38	37.91	50.97	54.09
				✓	71.22	20.75	72.95	26.15	55.32	39.64	48.92	56.38
✓					78.05	8.14	82.31	10.89	67.45	12.51	75.83	13.33
	✓				74.71	14.69	78.87	17.81	59.84	28.08	60.67	34.93
✓	✓				78.37	6.25	83.17	8.95	68.34	9.79	77.51	10.61
✓	✓	✓			78.88	5.06	83.54	8.54	69.48	6.51	78.32	8.97
✓	✓	✓	✓		79.63	4.63	84.24	7.30	71.48	7.26	81.33	8.73
✓	✓	✓	✓	✓	79.28	4.74	84.11	7.56	72.31	6.28	79.64	8.39
✓	✓	✓	✓	✓	80.06	4.70	85.18	5.82	74.91	6.99	82.75	8.27

D. Ablation Studies

In this subsection, we analyze the effects of the proposed components, various importance-aware orthogonal constraint strategies, and different levels of importance. The benchmarks used include two conventional CL benchmarks, one long-sequence CL benchmark, and one OOD CL benchmark: 10-split ImageNet-R (abbreviated as 10S-IN-R), 10-split DomainNet (10S-Domain), 50-split ImageNet-R and 9-split RESISC.

1) *Effects of different components*: There are five main components in our approach, including the orthogonal projection matrices $\mathbf{B}_1$ and $\mathbf{B}_2$, the invariant distribution loss $\mathcal{L}_{LN}$, the importance strategy represented by four masks $\mathbf{M}_{\{both,old,new,rest\}}$, and the classifier alignment. The effects of these components are presented in Table IV. Our findings are as follows:

- 1) The orthogonal projection for Affinity, *i.e.*, $\mathbf{B}_1$, is the crucial component for anti-forgetting. Only using $\mathbf{B}_1$ can increase accuracy by 12.6% and decrease forgetting by 23.0% in average. However, optimal performance can only be achieved when $\mathbf{B}_2$ and $\mathcal{L}_{LN}$ are collectively leveraged with $\mathbf{B}_1$, especially in the long-sequence and OOD CL scenarios.
- 2) The importance-aware orthogonal constraint strategy and the classifier alignment can further enhance the performance, where the former plays a more important role.

TABLE V
RESULTS OF DIFFERENT IMPORTANCE-AWARE ORTHOGONAL CONSTRAINT STRATEGIES. A DEFAULT UPDATE FORMULA AMONG (34) ~ (37) IS CHANGED AS ANOTHER FORMULA. a : $\Delta \mathbf{P}_{both} = [\eta \mathbf{B}_2 + (1 - \eta) \mathbf{I}] \mathbf{G}_P [\eta \mathbf{B}_1 + (1 - \eta) \mathbf{I}] \odot \mathbf{M}_{both}$; b : $\Delta \mathbf{P}_{old} = \mathbf{B}_2 \mathbf{G}_P \mathbf{B}_1 \odot \mathbf{M}_{old}$; c : $\Delta \mathbf{P}_{new} = \mathbf{B}_2 \mathbf{G}_P \mathbf{B}_1 \odot \mathbf{M}_{new}$; d : $\Delta \mathbf{P}_{rest} = \mathbf{B}_2 \mathbf{G}_P \mathbf{B}_1 \odot \mathbf{M}_{rest}$.

Strategy	10S-IN-R		10S-Domain		50S-IN-R		9S-RESISC	
	Acc.↑	Fgt.↓	Acc.↑	Fgt.↓	Acc.↑	Fgt.↓	Acc.↑	Fgt.↓
Default	80.06	4.70	85.18	5.82	74.91	6.99	82.75	8.27
a .	78.78	5.04	83.88	6.33	72.94	8.68	82.12	8.94
b .	79.27	4.98	84.35	6.05	73.31	7.34	81.91	8.58
c .	79.79	4.57	84.40	5.37	74.16	6.30	82.30	7.52
d .	79.44	4.38	84.76	5.21	73.77	6.09	81.68	7.21

Specifically, they improve the average accuracy by 1.62% and 1.28%, respectively. With the both techniques, accuracy can be enhanced by an average of 3.17% for our prior work VPT-NSP². Similarly, these two techniques exhibit more significant improvements (4.93% in average) under the challenging long-sequence CL and OOD CL settings.

- 3) Classifier alignment may not benefit continual learning when representations drift significantly. However, it proves effective for our approach, as it maintains representations with minimal drift. As can be seen, when equipped with the classifier alignment, the baseline shows slight degradation in performance (-1.17% in average). In contrast, as aforementioned, the classifier alignment brings 1.28% accuracy improvement for our approach. This suggests that the classifier alignment can achieve its maximum effectiveness within our framework owing to the nearly no drifting property of representations.

2) *Different update strategies*: We modify the importance-aware orthogonal constraint strategy in (34)~(37) to create four different strategies, each by replacing one update formula at a time. The modified update formulas and the results of the various strategies are shown in Table V. The meaning of each strategy is explained as follows. a : The orthogonal constraints for parameters important to both old and new tasks are changed from a strict version to a relaxed version. b : Parameters important to old tasks are updated using orthogonal projections instead of being kept frozen. c : Parameters important to new tasks are updated through orthogonal projections rather than without any constraints. d : The updates of the rest parameters are projected

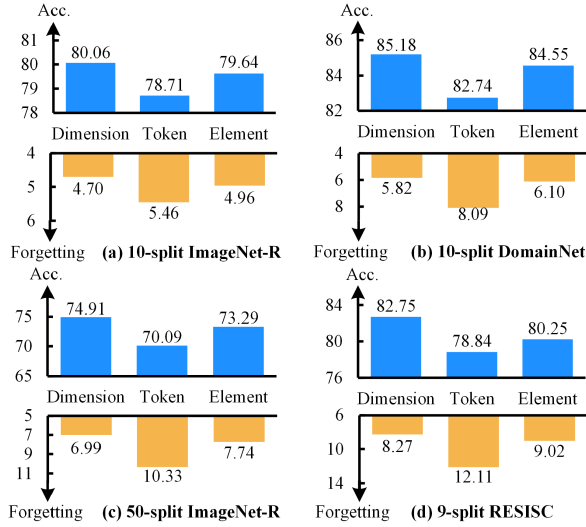


Fig. 5. Performance of different levels in terms of computing importance in the proposed importance-aware orthogonal constraint strategy.

using strict orthogonal matrices instead of relaxed orthogonal matrices.

The results in Table V demonstrate that our default strategy, *i.e.*, (34)~(37), is the optimal one among all the listed strategies. Comparing each variant with the default strategy, relaxing the orthogonality for $\Delta \mathbf{P}_{both}$ (*i.e.*, strategy *a*) and training $\Delta \mathbf{P}_{old}$ (*i.e.*, strategy *b*) can weaken the stability of the model, resulting in increased forgetting and lower accuracy. Although strategies *c* and *d* achieve lower forgetting by enforcing strict orthogonal constraints, the plasticity is limited and consequently the overall performance degrades. In summary, the proposed default strategy, which utilizes importance to guide parameter updates, can achieve a better balance between stability and plasticity.

3) *Different levels of importance*: Dimensional importance is employed in our approach to generate binary masks, as introduced in (28). We explore more different levels of importance, including a token level and an element level. The token level means that the parameters within each token share the same importance value. In this case, (28) is modified to $\bar{\mathbf{E}}_{\mathbf{P}_o} = [\sum_{j=1}^D \mathbf{E}_{\mathbf{P}_o}^{(1,j)}, \dots, \sum_{j=1}^D \mathbf{E}_{\mathbf{P}_o}^{(M,j)}] \in \mathbb{R}^M$. The element-level importance directly applies (27) to generate binary masks, *i.e.*, $\bar{\mathbf{E}}_{\mathbf{P}_o} = \mathbf{E}_{\mathbf{P}_o}$, where each element obtains an independent importance.

Fig. 5 displays the performance comparison of these three importance levels. The dimension-level importance achieves the highest accuracy, exceeding the token level and the element level by 3.13% and 1.29%, respectively. We argue that the token-level importance, where a single importance value is shared among 768 parameters, is overly coarse-grained. This can lead to many important elements within a token being misclassified as unimportant (and vice versa), resulting in relative poor performance. On the other hand, using element-level importance for parameter updates leads to inconsistent update strategies across the parameters within the same vector, thereby distorting the

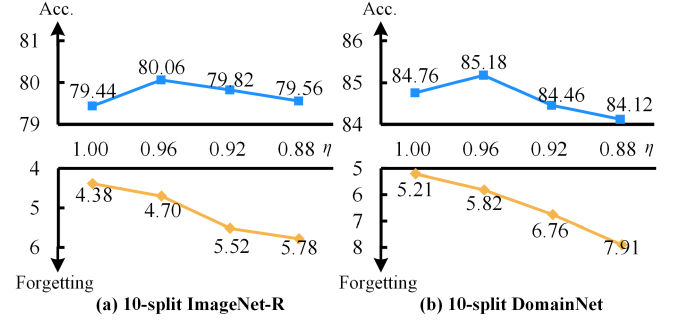


Fig. 6. Performance curves with regard to the relaxation parameter η in (37). A smaller value of η results in a greater degree of relaxation for the orthogonal projections.

vector's semantic meaning and constraining performance gains. In contrast, dimension-level importance strikes a better balance by preserving the vector's semantics while providing finer-grained importance calculation (with a single importance value covering 4 parameters), making it a more effective approach.

E. Effects of Hyper-Parameters

We experiment on 10-split ImageNet-R and 10-split DomainNet to evaluate the effects of key hyper-parameters, including the relaxation degree for orthogonality controlled by η in (37), the specified number of important dimensions (K in (29)), and the prompt depth L , *i.e.*, the number of ViT layers equipped with prompts.

1) *Relaxation degree*: We use the hyper-parameter η in (37) to relax the orthogonal projections. The effect of η on accuracy and forgetting is presented in Fig. 6. As η increases, accuracy improves first due to enhanced plasticity. Meanwhile, the stability is weakened and the forgetting steadily increases. After the balance point where the accuracy reaches the peak, the accuracy declines when η is further reduced. The reason is that the orthogonality is over relaxed, which results in serious forgetting. To summarize, our importance-aware orthogonal constraint strategy benefits from adjusting the relaxation degree, enabling a better trade-off between stability and plasticity and thereby improving overall performance.

2) *Top- K important dimensions*: In the importance-aware orthogonal constraint strategy, the dimensions corresponding to the top- K importance values (out of a total of $D=768$ dimensions) in (29) are identified as important for the old or new task. We vary K from 10 to 150 in increments of 50 (except for the first increment), and the impact of K on performance is shown in Fig. 7. As K increases, accuracy initially improves but eventually declines, while forgetting continues to increase. This behavior is similar to the trend observed with the parameter η . These results suggest that varying K provides an alternative method for balancing stability and plasticity.

3) *Depth and length of the prompt*: The performance curves of VPT-Seq and VPT-NSP²⁺⁺ with respect to prompt depth (L) and prompt length (M) are displayed in Fig. 8 and Fig. 9, respectively. As prompts are inserted into more and deeper layers,

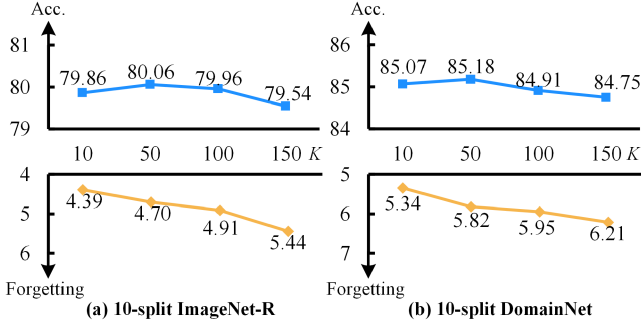


Fig. 7. Performance curves in terms of K in (29). K denotes the specified number of important values in the initial importance matrices $\mathbf{E}_{\mathbf{P}_o}$ and $\mathbf{E}_{\mathbf{P}_n}$.

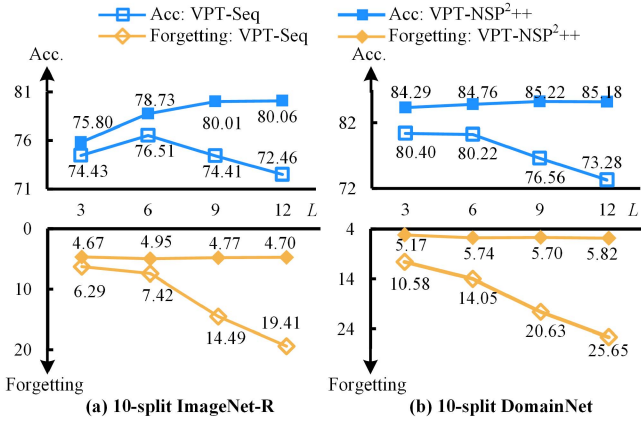


Fig. 8. Effect of prompt depth (denoted as L , i.e., the number of layers incorporating prompts from the first to the last) on the baseline VPT-Seq and our approach VPT-NSP²⁺⁺.

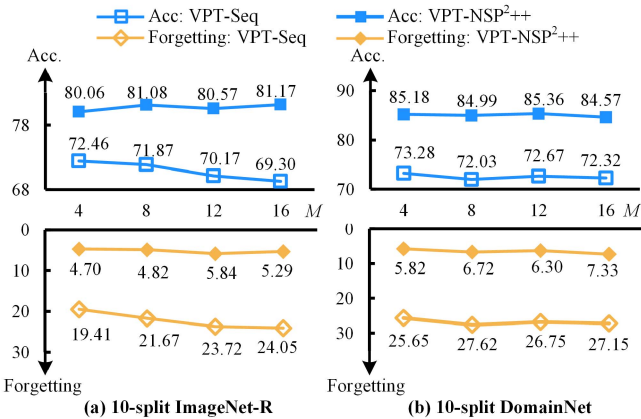


Fig. 9. Effect of prompt length (i.e., M) on the baseline VPT-Seq and our approach VPT-NSP²⁺⁺.

the performance of the baseline initially improves but eventually declines sharply, accompanied by a significant increase in forgetting. The initial accuracy improvement is due to enhanced plasticity, while the subsequent decline is attributed to reduced stability. In contrast, VPT-NSP²⁺⁺ effectively capitalizes on the increased adaptability to achieve superior performance while

TABLE VI
SENSITIVITY OF THE MODEL TO THE NUMBER OF IMPORTANT DIMENSIONS K . THE RESULTS WITH THE DEFAULT VALUES ARE UNDERLINED, AND THE BEST RESULTS ARE SHOWN IN BOLD.

K	50S-ImageNet-R		9S-RESISC	
	Acc.	Forgetting	Acc.	Forgetting
30	74.63	6.75	82.68	8.16
40	74.79	6.90	82.88	8.33
<u>50</u>	<u>74.91</u>	6.99	82.75	8.27
60	74.95	7.12	82.70	8.39
70	74.86	7.27	82.64	8.67
100	74.50	7.58	82.13	9.04
200	73.44	8.39	81.29	9.74

TABLE VII
SENSITIVITY OF THE MODEL TO THE RELAXATION PARAMETER η . THE RESULTS WITH THE DEFAULT VALUES ARE UNDERLINED, AND THE BEST RESULTS ARE SHOWN IN BOLD.

η	50S-ImageNet-R		9S-RESISC	
	Acc.	Forgetting	Acc.	Forgetting
0.99	74.74	6.05	82.33	7.88
0.98	74.90	6.17	82.57	7.92
0.97	75.22	6.58	82.65	8.05
<u>0.96</u>	<u>74.91</u>	6.99	82.75	8.27
0.95	74.68	7.32	82.60	8.53
0.9	73.75	8.17	82.47	8.80
0.8	72.11	9.59	81.85	9.43

maintaining low forgetting. The prompt length has a moderate effect on performance, with the accuracy of VPT-NSP²⁺⁺ slightly improving as M increases. Nevertheless, we uniformly set $M = 4$ across all experiments for consistency.

4) *Discussion on hyper-parameter selection:* In practice, the values of K and η can be selected from a reasonable range ($K \in [30, 70]$ and $\eta \in [0.95, 0.99]$), as our model shows little sensitivity to these hyper-parameters. To support this, we conduct experiments in challenging long-sequence and OOD scenarios, which closely reflect practical applications.

As shown in the Table VI and Table VII, varying η in the range $[0.95, 0.99]$ and K in the range $[30, 70]$ results in only minor changes in accuracy. Specifically, in these ranges, the accuracy on the 50S-ImageNet-R benchmark fluctuates within 0.5% (from 74.68% to 75.22%), and on the 9S-RESISC benchmark within 0.4% (from 82.33% to 82.75%). Similarly, varying K from 30 to 70 leads to accuracy changes within 0.3% on both benchmarks. This demonstrates that the performance remains stable when the parameters are set within these reasonable intervals. In practice, we recommend setting $\eta=0.96$ and $K=50$ as default values, as they are sufficient to achieve a good balance between stability and plasticity. These values have been consistently used in all reported experiments and can be directly adopted for new datasets or tasks without further tuning.

F. Model Analysis

1) *Evolution of accuracy and loss:* To better understand the anti-forgetting effect of the proposed approach, we visualize the task-by-task accuracy curves and the training loss of old

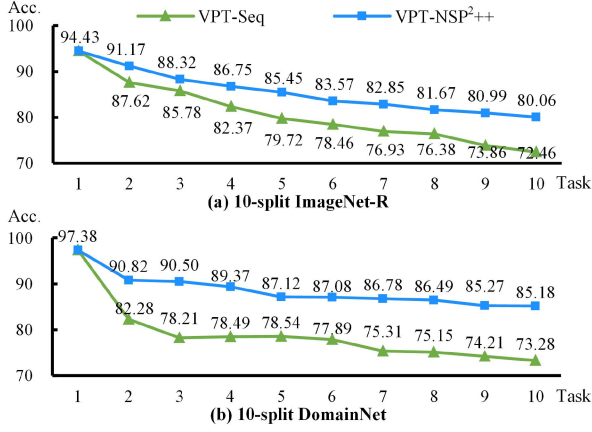


Fig. 10. Task-by-task accuracy changing curves of VPT-Seq and VPT-NSP²⁺⁺ on two benchmarks.

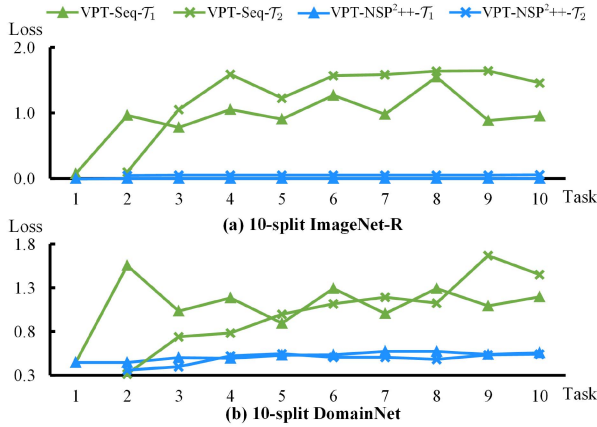


Fig. 11. Training loss curves of VPT-Seq and VPT-NSP²⁺⁺ on tasks T_1 and T_2 . Each point represents the loss value on the training data of task T_1/T_2 when the model finishes learning the t -th task.

tasks. As displayed in Fig. 10, both the baseline VPT-Seq and our approach VPT-NSP²⁺⁺ start with the same accuracy on the first task. However, as the number of new tasks increases, the accuracy of VPT-NSP²⁺⁺ consistently remains higher than that of VPT-Seq. Furthermore, the performance gap between the two approaches widens over time, highlighting the superior anti-forgetting capability of our method.

The evolution of training loss is shown in Fig. 11. After the model completes training on a subsequent task, it is evaluated on the training samples from two previously learned tasks, T_1 and T_2 , to compute the respective loss values. If the representations of the old data remain stable, the loss values on these tasks should remain unchanged. It is clear to see that the loss of VPT-Seq on old tasks rises and fluctuates heavily. In contrast, the loss of VPT-NSP²⁺⁺ on the two previously learned tasks remains almost constant at a low level. This demonstrates that our approach effectively prevents the drift of representations for previous tasks, thereby exhibiting strong anti-forgetting capabilities.

TABLE VIII

RESOURCE CONSUMPTION OF FOUR METHODS ON 10-SPLIT IMAGENET-R. EXTRA MEMORY (IN MEGABYTE, MB): THE ADDITIONALLY OCCUPIED GPU MEMORY COMPARED WITH THE BASELINE. TRAINING TIME (IN MILLISECOND, MS): THE TIME OF FORWARD AND BACKWARD PROPAGATIONS FOR TRAINING EACH SAMPLE. INFERENCE GFLOPS AND INFERENCE TIME (IN MS): THE GIGA FLOPS AND THE FORWARD PROPAGATION TIME FOR INFERRING ONE IMAGE.

Method	Training stage		Inference Stage		Acc.
	Extra memory(MB)	Training time (ms)	Inference GFLOPs	Inference time (ms)	
CODA-Prompt [20]	15.36	4.78	70.35	2.41	75.45
CPrompt [25]	3.68	7.21	73.98	2.48	77.14
VPT-Seq	0	2.05	35.45	0.85	72.46
VPT-NSP²⁺⁺	57.07	2.15	35.45	0.85	80.06

2) *Resource consumption*: We analyze the training and inference resource consumption of our approach and the baseline on 10-split ImageNet-R, as shown in Table VIII. Additionally, the resource consumption of two existing prompt-tuning-based methods, CODA-Prompt [20] and CPrompt [25], is also reported for comparison. Note that we use the same hardware (a single NVIDIA GeForce RTX 4090 GPU) for all the approaches. Although our approach requires more memory than the two methods, the extra memory (56.81 MB) is negligible since it accounts for only approximately 0.24% of the GPU memory (24 GB). In terms of training time, inference overhead, inference time and accuracy, our approach demonstrates a clear advantage. Especially, the inference overhead and time of VPT-NSP²⁺⁺ are the same as that of a single fine-tuned ViT (*i.e.*, the baseline VPT-Seq), as our approach does not rely on a query function implemented by another ViT. The low latency of VPT-NSP²⁺⁺ highlights its significant potential for real-world applications.

The detailed analysis of extra training memory and complexity are introduced as follows. *Extra training memory*: Our approach introduces the UCMs $\mathbf{C}_1 \in \mathbb{R}^{D \times D}$, $\mathbf{C}_2 \in \mathbb{R}^{M \times M}$, projection matrices $\mathbf{B}_1 \in \mathbb{R}^{D \times D}$, $\mathbf{B}_2 \in \mathbb{R}^{M \times M}$, distribution vectors $\mu_{P_t}, \sigma_{P_t} \in \mathbb{R}^M$, importance matrices $\mathbf{E}_{P_o}, \mathbf{E}_{P_n} \in \mathbb{R}^{M \times D}$ and masks $\mathbf{M}_{\{both, old, new, rest\}} \in \mathbb{R}^D$. The total number of extra parameters is $2(D^2 + M^2 + MD + 2D + M)L$, where L is the prompt depth. Although the recorded distributions for classifier alignment also need to be memorized, it is worth noting that they do not occupy GPU memory. *Extra training complexity*: We neglect the computation required only once per task, such as the SVD on UCMs. The main extra training overhead is raised by the projections and masks performed on gradients, as formulated in (34)~(37). The computational overhead of the orthogonal projections in (34) and (37) is $\mathcal{O}(2LDM(D + M))$. For the mask operations in all the four formulas, the overhead is $\mathcal{O}(4LDM)$. Overall, the main extra complexity is $\mathcal{O}(LDM(D + M + 2))$.

VI. CONCLUSION

In this work, we introduced an importance-aware orthogonal regularization method for mitigating catastrophic forgetting in Vision Transformers. We proposed two orthogonality conditions along with a constraint of invariant prompt distribution, which

can ensure that prompt updates for new tasks preserve the knowledge learned from previous tasks. To achieve these conditions, we implemented a null-space projection solution and incorporated an invariant prompt distribution loss during optimizing the prompts in the new task. Additionally, the proposed importance-aware orthogonal constraint strategy enables the model to effectively balance stability and plasticity. Extensive experiments on multiple class-incremental learning benchmarks demonstrate the superiority of our approach over existing state-of-the-art methods. Furthermore, our method exhibits strong generalizability in long-sequence and out-of-distribution continual learning scenarios. Future work will explore further improvements in scalability and the application to other Transformer-based architectures across diverse continual learning scenarios.

REFERENCES

- [1] S. Dohare, J. F. Hernandez-Garcia, Q. Lan, P. Rahman, A. R. Mahmood, and R. S. Sutton, "Loss of plasticity in deep continual learning," *Nature*, vol. 632, no. 8026, pp. 768–774, Aug. 2024.
- [2] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "iCaRL: Incremental classifier and representation learning," in *Proc. Int. Conf. Comput. Vis.*, 2017, pp. 2001–2010.
- [3] D.-W. Zhou, Q.-W. Wang, H.-J. Ye, and D.-C. Zhan, "A model or 603 exemplars: Towards memory-efficient class-incremental learning," in *Proc. Int. Conf. Learn. Representations*, 2023, pp. 9689–9723.
- [4] R. Aljundi, M. Lin, B. Goujaud, and Y. Bengio, "Gradient based sample selection for online continual learning," in *Proc. Neural Inf. Process. Syst.*, 2019, pp. 11816–11825.
- [5] S. Yan, J. Xie, and X. He, "DER: Dynamically expandable representation for class incremental learning," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 3014–3023.
- [6] Q. Gao, Z. Luo, D. Klabjan, and F. Zhang, "Efficient architecture search for continual learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8555–8565, Nov. 2023.
- [7] J. Yoon, S. Kim, E. Yang, and S. J. Hwang, "Scalable and order-robust continual learning with additive parameter decomposition," in *Proc. Int. Conf. Learn. Representations*, 2020, pp. 9728–9742.
- [8] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.
- [9] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. Int. Conf. Mach. Learn.*, 2017, vol. 70, pp. 3987–3995.
- [10] G. Zeng, Y. Chen, B. Cui, and S. Yu, "Continual learning of context-dependent processing in neural networks," *Nature Mach. Intell.*, vol. 1, no. 8, pp. 364–372, Aug. 2019.
- [11] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 611–631.
- [12] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-End object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, vol. 12346, pp. 213–229.
- [13] A. Kirillov et al., "Segment anything," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 3992–4003.
- [14] M. Jia et al., "Visual prompt tuning," in *Proc. Eur. Conf. Comput. Vis.*, 2022, vol. 13693, pp. 709–727.
- [15] S. Chen et al., "AdaptFormer: Adapting vision transformers for scalable visual recognition," in *Proc. Neural Inf. Process. Syst.*, 2022, pp. 16664–16678.
- [16] X. L. Li and P. Liang, "Prefix-Tuning: Optimizing continuous prompts for generation," in *Proc. Annu. Meeting Assoc. Comput. Linguist.*, 2021, pp. 4582–4597.
- [17] Z. Wang et al., "Learning to prompt for continual learning," in *Proc. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 139–149.
- [18] Z. Wang et al., "Dual Prompt: Complementary prompting for rehearsal-free continual learning," in *Proc. Eur. Conf. Comput. Vis.*, 2022, vol. 13686, pp. 631–648.
- [19] Y. Wang, Z. Huang, and X. Hong, "S-Prompts learning with pre-trained transformers: An Occam's razor for domain incremental learning," in *Proc. Neural Inf. Process. Syst.*, 2022, pp. 5682–5695.
- [20] J. S. Smith et al., "CODA-Prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning," in *Proc. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 11909–11 919.
- [21] M. G. Z. A. Khan, M. F. Naeem, L. Van Gool, D. Stricker, F. Tombari, and M. Z. Afzal, "Introducing language guidance in prompt-based continual learning," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 11429–11 439.
- [22] Y.-M. Tang, Y.-X. Peng, and W.-S. Zheng, "When prompt-based incremental learning does not meet strong pretraining," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 1706–1716.
- [23] L. Wang, J. Xie, X. Zhang, M. Huang, H. Su, and J. Zhu, "Hierarchical decomposition of prompt-based continual learning: Rethinking obscured sub-optimality," in *Proc. Neural Inf. Process. Syst.*, 2023, pp. 69054–69076.
- [24] R. Wang et al., "AttriCLIP: A non-incremental learner for incremental knowledge learning," in *Proc. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 3654–3663.
- [25] Z. Gao, J. Cen, and X. Chang, "Consistent prompting for rehearsal-free continual learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 28463–28 473.
- [26] D. Kim et al., "One size fits all for semantic shifts: Adaptive prompt tuning for continual learning," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 24658–24673.
- [27] M. R. Kurniawan et al., "Evolving parameterized prompt memory for continual learning," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 13301–13309.
- [28] Z. Li et al., "Steering prototypes with prompt-tuning for rehearsal-free continual learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2024, pp. 2523–2533.
- [29] A. Roy, R. Moulick, V. K. Verma, S. Ghosh, and A. Das, "Convolutional prompting meets language models for continual learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 23616–23 626.
- [30] A. Vaswani et al., "Attention is all you need," in *Proc. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [31] L. J. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
- [32] S. Wang, X. Li, J. Sun, and Z. Xu, "Training networks in null space of feature covariance for continual learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2021, pp. 184–193.
- [33] Y. Lu et al., "Visual prompt tuning in null space for continual learning," in *Proc. Neural Inf. Process. Syst.*, 2024, pp. 7878–7901.
- [34] R. Zhu et al., "ScratchDet: Training single-shot object detectors from scratch," in *Proc. Comput. Vis. Pattern Recognit.*, 2019, pp. 2268–2277.
- [35] K. He, R. B. Girshick, and P. Dollár, "Rethinking imagenet pre-training," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 4917–4926.
- [36] Z. Shen, Z. Liu, J. Li, Y.-G. Jiang, Y. Chen, and X. Xue, "DSOD: Learning deeply supervised object detectors from scratch," in *Proc. Int. Conf. Comput. Vis.*, 2017, pp. 1937–1945.
- [37] S.-A. Rebuffi, H. Bilen, and A. Vedaldi, "Learning multiple visual domains with residual adapters," in *Proc. Neural Inf. Process. Syst.*, 2017, pp. 506–516.
- [38] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. Mach. Learn.*, 2019, vol. 97, pp. 2790–2799.
- [39] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Representations*, 2022, pp. 12513–12525.
- [40] Y. Xing et al., "Dual modality prompt tuning for vision-language pre-trained model," *IEEE Trans. Multimedia*, vol. 26, pp. 2056–2068, 2024.
- [41] Q. Gao et al., "A unified continual learning framework with general parameter-efficient tuning," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 11449–11459.
- [42] G. Zhang, L. Wang, G. Kang, L. Chen, and Y. Wei, "SLCA: Slow learner with classifier alignment for continual learning on a pre-trained model," in *Proc. Int. Conf. Comput. Vis.*, Oct. 2023, pp. 19091–19101.
- [43] B. Bowman et al., "À-la-carte prompt tuning (APT): Combining distinct data via composable prompting," in *Proc. Comput. Vis. Pattern Recognit.*, 2023, pp. 14984–14 993.
- [44] Y.-S. Liang and W.-J. Li, "InFLoRA: Interference-free low-rank adaptation for continual learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 23638–23 647.
- [45] D.-W. Zhou, H.-L. Sun, H.-J. Ye, and D.-C. Zhan, "Expandable sub-space ensemble for pre-trained model-based class-incremental learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 23554–23 564.
- [46] J. Yu et al., "Boosting continual learning of vision-language models via mixture-of-experts adapters," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 23219–23 230.

- [47] M. Sandler, A. Zhmoginov, M. Vladymyrov, and A. Jackson, "Fine-tuning image transformers using learnable memory," in *Proc. Comput. Vis. Pattern Recognit.*, 2022, pp. 12145–12 154.
- [48] Y. Wang, Z. Ma, Z. Huang, Y. Wang, Z. Su, and X. Hong, "Isolation and impartial aggregation: A paradigm of incremental learning without interference," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 10209–10217.
- [49] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, and A. van den Hengel, "RanPAC: Random projections and pre-trained models for continual learning," in *Proc. Neural Inf. Process. Syst.*, 2023, pp. 12022–12053.
- [50] W.-C. Huang, C.-F. Chen, and H. Hsu, "OVOR: OnePrompt with virtual outlier regularization for rehearsal-free class-incremental learning," in *Proc. Int. Conf. Learn. Representations*, 2024, pp. 39355–39374.
- [51] A. Chaudhry, N. Khan, P. K. Dokania, and P. H. S. Torr, "Continual learning in low-rank orthogonal subspaces," in *Proc. Neural Inf. Process. Syst.*, 2020, pp. 9900–9911.
- [52] G. Saha, I. Garg, and K. Roy, "Gradient projection memory for continual learning," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 944–961.
- [53] Y. Kong, L. Liu, Z. Wang, and D. Tao, "Balancing stability and plasticity through advanced null space in continual learning," in *Proc. Eur. Conf. Comput. Vis.*, 2022, vol. 13686, pp. 219–236.
- [54] Y. Hu, D. Cheng, D. Zhang, N. Wang, T. Liu, and X. Gao, "Task-aware orthogonal sparse network for exploring shared knowledge in continual learning," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 19153–19164.
- [55] S. A. Shah, F. Palmieri, and M. Datum, "Optimal filtering algorithms for fast learning in feedforward neural networks," *Neural Netw.*, vol. 5, no. 5, pp. 779–787, 1992.
- [56] D. Deng, G. Chen, J. Hao, Q. Wang, and P.-A. Heng, "Flattening sharpness for dynamic gradient projection memory benefits continual learning," in *Proc. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 18710–18721.
- [57] Z. Zhao et al., "Rethinking gradient projection continual learning: Stability/Plasticity feature space decoupling," in *Proc. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 3718–3727.
- [58] S. Wang, X. Li, J. Sun, and Z. Xu, "Training networks in null space of feature covariance with self-supervision for incremental learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2563–2580, Apr. 2025.
- [59] C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*. Philadelphia, PA, USA: Society for Industrial and Applied Mathematics (SIAM), 2000.
- [60] G. M. van de Ven, T. Tuytelaars, and A. S. Tolias, "Three types of incremental learning," *Nature Mach. Intell.*, vol. 4, no. 12, pp. 1185–1197, 2022.
- [61] M. Masana, X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. van de Weijer, "Class-Incremental learning: Survey and performance evaluation on image classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5513–5533, May 2023.
- [62] D.-W. Zhou, Q.-W. Wang, Z.-H. Qi, H.-J. Ye, D.-C. Zhan, and Z. Liu, "Class-Incremental learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9851–9873, Dec. 2024.
- [63] H. He, J. Cai, J. Zhang, D. Tao, and B. Zhuang, "Sensitivity-Aware visual parameter-efficient fine-tuning," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 11791–11 801.
- [64] Z. Zhang et al., "Gradient-based parameter selection for efficient fine-tuning," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 28566–28 577.
- [65] P. Khosla et al., "Supervised contrastive learning," in *Proc. Neural Inf. Process. Syst.*, 2020, pp. 18661–18673.
- [66] Y. Tan, Q. Zhou, X. Xiang, K. Wang, Y. Wu, and Y. Li, "Semantically-Shifted incremental adapter-tuning is a continual ViTransformer," in *Proc. Comput. Vis. Pattern Recognit.*, 2024, pp. 23252–23262.
- [67] L. Yu et al., "Semantic drift compensation for class-incremental learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2020, pp. 6980–6989.
- [68] D. Hendrycks et al., "The many faces of robustness: A critical analysis of out-of-distribution generalization," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 8320–8329.
- [69] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," Univ. Toronto, Toronto, ON, Canada, Tech. Rep., 2009. [Online]. Available: <https://www.cs.toronto.edu/kriz/learning-features-2009-TR.pdf>
- [70] X. Peng, Q. Bai, X. Xia, Z. Huang, K. Saenko, and B. Wang, "Moment matching for multi-source domain adaptation," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 1406–1415.
- [71] Y. Zhang, Z. Yin, J. Shao, and Z. Liu, "Benchmarking omni-vision representation through the lens of visual realms," in *Proc. Eur. Conf. Comput. Vis.*, 2022, vol. 13667, pp. 594–611.
- [72] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, and Z. Liu, "Revisiting class-incremental learning with pre-trained models: Generalizability and adaptivity are all you need," *Int. J. Comput. Vis.*, vol. 133, no. 3, pp. 1012–1032, 2025.
- [73] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2217–2226, Jul. 2019.
- [74] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
- [75] O. Russakovsky et al., "Imagenet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, 2015.
- [76] E. D. Cubuk, B. Zoph, D. Mané, V. Vasudevan, and Q. V. Le, "AutoAugment: Learning augmentation strategies from data," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 113–123.
- [77] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Representations*, 2015.
- [78] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Neural Inf. Process. Syst.*, 2019, pp. 8024–8035.
- [79] R. Wightman, "Pytorch image models," 2019. [Online]. Available: <https://github.com/rwightman/pytorch-image-models>
- [80] T. D. Min, M. Mancini, K. Alahari, X. Alameda-Pineda, and E. Ricci, "On the effectiveness of layernorm tuning for continual learning in vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops*, 2023, pp. 3577–3586.
- [81] X. Gao, S. Dong, Y. He, Q. Wang, and Y. Gong, "Beyond prompt learning: Continual adapter for efficient rehearsal-free continual learning," in *Proc. Eur. Conf. Comput. Vis.*, 2024, vol. 15143, pp. 89–106.
- [82] Y. Kim, Y. Li, and P. Panda, "One-stage prompt-based continual learning," in *Proc. Eur. Conf. Comput. Vis.*, 2024, vol. 15071, pp. 163–179.
- [83] D. Li, T. Wang, J. Chen, Q. Ren, K. Kawaguchi, and Z. Zeng, "Towards continual learning desiderata via HSIC-bottleneck orthogonalization and equiangular embedding," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 13464–13473.
- [84] J. Qiao et al., "Prompt gradient projection for continual learning," in *Proc. Int. Conf. Learn. Representations*, 2024, pp. 3856–3877.
- [85] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 9630–9640.
- [86] T. Ridnik, E. B. Baruch, A. Noy, and L. Zelnik, "ImageNet-21K Pretraining for the Masses," in *Proc. Neural Inf. Process. Syst. Datasets Benchmarks*, 2021, pp. 1161–1172.
- [87] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, vol. 139, pp. 8748–8763.
- [88] M. Cherti et al., "Reproducible scaling laws for contrastive language-image learning," in *Proc. Comput. Vis. Pattern Recognit.*, 2023, pp. 2818–2829.



Shizhou Zhang received the BE and PhD degrees from Xi'an Jiaotong University, Xi'an, China, in 2010 and 2017, respectively. He is currently an associate professor (Tenured) with Northwestern Polytechnical University, Xi'an. His research interests include content-based image analysis, pattern recognition, and machine learning, specifically in vision tasks under continual learning and open world settings, such as image classification, object detection, and reidentification.



Yue Lu received the BE degree in automation from Beijing Information Science and Technology University, Beijing, China, in 2020, and the ME degree in computer science in 2023 from Northwestern Polytechnical University, Xi'an, China, where he is currently working toward the PhD degree in computer science with the School of Computer Science. His current research interests include computer vision and deep learning, especially continual learning.



De Cheng received the BS and PhD degrees from Xi'an Jiaotong University, Xi'an, China, in 2011 and 2017, respectively. He is currently an associate professor with the School of Telecommunications Engineering, Xidian University, Xi'an. From 2015 to 2017, he was a visiting scholar with Carnegie Mellon University, Pittsburgh, PA, USA. His research interests include pattern recognition, machine learning, and multimedia analysis.



Peng Wang (Member, IEEE) received the bachelor's degree in electrical engineering and automation and the PhD degree in control science and engineering from Beihang University, Beijing, China, in 2004 and 2011, respectively. He was with the School of Computer Science, University of Adelaide, Adelaide, SA, Australia, for four years. He is currently a professor with the School of Computer Science, Northwestern Polytechnical University, Xi'an, China. His research interests include computer vision, machine learning, and artificial intelligence.



Yinghui Xing (Member, IEEE) received the BS and PhD degrees from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2014 and 2020, respectively. She is currently an associate professor with the School of Computer Science, Northwestern Polytechnical University, Xi'an. Her research interests include remote sensing image processing, image fusion, and image super resolution.



Nannan Wang (Senior Member, IEEE) received the BSc degree in information and computation science from the Xi'an University of Posts and Telecommunications, in 2009, and the PhD degree in information and telecommunications engineering from Xidian University, in 2015. From 2011 to 2013, he was a visiting PhD student with the University of Technology Sydney, NSW, Australia. He is currently a professor with the State Key Laboratory of Integrated Services Networks, Xidian University. His current research interests include computer vision and machine learning.



Yanning Zhang (Fellow, IEEE) received the BE degree from the Dalian University of Technology, Dalian, China, in 1988, and the PhD degree from the School of Marine Engineering, Northwestern Polytechnical University, Xi'an, China, in 1996. She is currently a professor and the executive dean with the School of Computer Science, Northwestern Polytechnical University. Her research interests include computer vision and pattern recognition, image and video processing, and intelligent information processing.